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Katie Berns (katie@dsv.su.se)
Stockholm University
Chiara Rossitto
Stockholm University
Jakob Tholander
Stockholm University

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Learning from Other Communities: Organising Collective Action in a Grassroots Food-sharing Initiative

Katie Berns[1*], Chiara Rossitto and Jakob Tholander

Department of System and Computer Sciences, Stockholm University, Borgarfjordsgatan 12, Kista, 16407, Sweden.

*Corresponding author(s). E-mail(s): katie@dsv.su.se;
Contributing authors: chiara@dsv.su.se; jakob@dsv.su.se;

Abstract

This paper illustrates the work of creating, infrastructuring, and organising a food-sharing community from the ground up. Drawing on Participatory Action Research (PAR) and long-term engagement with FoodSharing Stockholm, the paper shows how the processes of starting up a grassroots initiative are shaped by participants' direct experience and knowledge of other related initiatives. The analysis draws attention to: (1) how central activities such as recruiting volunteers, choosing digital tools, and establishing partnerships with food donors are conceived and organised, (2) the concrete challenges of sharing surplus food, such as adopting a distribution model and negotiating fairness, and (3) how governance and decision-making models are adopted and (re)negotiated over time. The paper shows the impact that previous experiences can have on communities' efforts get up and running and re-orients design visions towards the appropriation of existing sociotechnical practices, rather than digital innovations. The paper continues the cycle of learning by contributing practical guidelines to build, sustain, and infrastructure surplus food-sharing initiatives where configuring capacities, configuring sociotechnical practices, and configuring participation, are discussed as central matters.

Keywords: Activism, Infrastructuring, Food-sharing, PAR, Grassroots

1 Introduction

Over the last decade, CSCW scholarship has investigated the role digital technology plays in structuring social movements and their efforts to organise collective action to raise awareness, promote social change, or influence policy. Sometimes at the intersection with HCI research, work in this area has outlined, for instance, how digital technology is adopted or can be tailored to structure participants' work practices and key values (Berns, Rossitto, & Tholander, 2021a; Ghoshal et al., 2019; Lu, 2021; Rossitto, Korsgaard, Lampinen, & Bødker, 2021; Rossitto & Lampinen, 2018; Voida, Yao, & Korn, 2015), develop local cooperative relations (Light & Miskelly, 2019; Mosconi et al., 2017), infrastructure the formation of publics (Le Dantec & DiSalvo, 2013), or to support different phases of community organising (Biørn-Hansen & Håkansson, 2018; Bødker, Korsgaard, & Saad-Sulonen, 2016; Svenfelt & Zapico, 2016).

Overall, these studies unravel the variety of processes and sociotechnical practices that make collective initiatives work, along with the attachments and commitments participants develop with each other and with specific matters of concern. They, however, say relatively little about the everyday practical work whereby such initiates are formed, get started, and become communities. This paper expands understandings of how organising and infrastructing unfold in practice, particularly through the relations that community initiatives have, or might develop, with related collectives. This paper shows that drawing on other communities' success stories, strategies to organise key practices, or framing shared problems are central to how goals, visions, and practical labour are defined, planned, and carried out. As illustrated by diverse examples – e.g, see FridaysForFuture (Brünker, Deitelhoff, & Mirbabaie, 2019), #MeToo (Hansson, Sveningsson, & Ganetz, 2021; Rossitto, Berns, & Larsen-Ledet, 2020), or #BlackLivesMatter (Mundt, Ross, & Burnett, 2018) – social movements and civic initiatives seldom evolve as isolated happenings. On the contrary, they draw from each other in framing calls for action, enabling forms of participation, defining main visions and goals, and in the type of outcomes and impact to strive for (Lampinen et al., 2022; Rossitto, 2021).

We introduce the concept of *"Collective histories of organising"* to account for the many and diverse previous experiences that volunteers bring to processes of setting up and maintaining a community-led initiative. The notion entails the previous knowledge, the direct or indirect experience people have, but also the sense-making processes and situated circumstances (e.g., recounting, discussing, negotiating, exemplifying) whereby individual histories become collectively relevant. We argue that outlining the explicit and more tacit interconnections that initiatives striving for social change develop with each other has a twofold relevance for CSCW. First, it shows the impact that (otherwise invisible) previous experiences can have on communities' efforts to articulate work (Star & Strauss, 1999), scope and shape the specific cooperative practices that characterise it. Second, it re-orients design visions towards the *adaptability* and *adoptability* of existing sociotechnical practices that community-led

initiatives can learn from each other, instead of always prioritising digital innovation as the end goal of design interventions. Resonating with previous studies that have emphasised civic initiatives' concerns for social change, rather than digital advances (Le Dantec et al., 2011; Schrock, 2018), and their expert, yet non-professional, design ability to create social alternatives (Manzini & Coad, 2015), our contribution bears important consequences for researchers and activist alike, as we outline both analytical and practical sensitivities to organise work and set up community activism from the ground up. As voluntary work often unfolds amid limited resources (e.g., human, physical, economic, infrastructural assets), we point to the relevance of existing framings and sociotechnical practices in helping community-led initiatives get started and sustain their work and commitment to it.

Narrowing down our focus to a specific domain, we investigate the start up of a grassroots food-sharing community operating in Stockholm, Sweden. Although heterogeneous in their ways of organising and contenting with food problems (A. Davies, 2019; Schanes & Stagl, 2019), food-sharing initiatives (Berns et al., 2021a; Chies, 2017; Ganglbauer, Fitzpatrick, Subasi, & Güldenpfennig, 2014) can be regarded as part of a global anti-food waste movement (Feng, Marek, & Tosun, 2022; Gollnhofer & Boller, 2020; Tartiu & Morone, 2017), a variegated activist reaction to the inefficiencies and inequalities of mainstream, capitalist food systems, particularly in the distribution chain (Gustavsson, Cederberg, & Sonesson, 2011; United Nations, 2013). Initiates within this movement are focused on raising awareness on the issues of food waste through providing alternatives to surplus food re-distribution.

The empirical material stems from a three year Participatory Action Research (PAR) approach (Hayes, 2014) focused on Foodsharing Stockholm, a grassroots community, where the first author has been an active member since its inception. The data has been collected by the first author through participant observation, from September 2019 to September 2022 [1]. The data consists of field notes from volunteer meetings, food collections and distributions, as well as from informal conversations with various community members.

The findings centre around three key themes illustrating how a food-sharing community is organised and infrastructured through learning from other communities and through the appropriation of various digital technologies. The first unpacks the role of previous experiences in articulating key visions as part of a strategy to initially set-up the community, create a digital infrastructure, recruit volunteers, and establish partnerships with food donors. The second looks at theme focuses on how organisers looked to similar communities for inspiration on existing food-sharing models, as well, as the negotiation of values such as consideration of fairness, and the effective dimensions that frame the distribution of surplus food. Finally, the third theme looks at the challenges connected to governing and maintaining the community. Here we unpack the community's decision to deviate from the approach of similar organisations

[1] There was a brief interruption in activity from the 25th of March to the 7th of July 2020 while the community adjusted/adapted operations to comply to safety recommendations and restriction put in place due to Covid19.

and adopt of non-hierarchical, consensus model of decision-making, leading to conflicts with practical concerns to get work done, as well as, challenges with regards to fluctuating participant numbers.

The paper introduces three themes, namely configuring capacities, configuring sociotechnical practices, and configuring participation, as practical guidelines that other initiatives can benefit from when organising. This is relevant to CSCW research as it opens up opportunities for scholars to co-operate with initiatives concerned with social change, collaboratively inform the design of sociotechnical practices, and identify recurrent challenges, processes and problems of technology appropriation that are relevant for infrastructuring community-led initiatives.

2 Related Work

In what follows, we first introduce previous CSCW and HCI scholarship on food-sharing initiatives. Situating our contribution within a broader research area, we then discuss related work that has investigated the processes whereby collective initiatives and social movements organise their actions both operationally and through governance and decision-making models.

2.1 Food-sharing as a collective action against food-waste

Waste of edible foodstuff results in an enormous loss of resources with long-lasting detrimental effects on global nutrition, carbon emissions and economic budgeting (Gustavsson et al., 2011). At the same time, it is a highly complex problem involving issues at macro level (i.e. policies, laws, and regulations) meso level (lack of infrastructure, confusion around food date labelling) and micro level (i.e. retail logistic and consumer behaviour) that require multiple types of interventions at different levels of the food supply chain (Aschemann-Witzel, De Hooge, Amani, Bech-Larsen, & Oostindjer, 2015; Falcone & Imbert, 2017; HLPE, 2014).

Over the last decade, HCI and CSCW scholarship has investigated the role technology can play in enabling just (Dombrowski, Brubaker, Hirano, Mazmanian, & Hayes, 2013; Prost, Crivellaro, Haddon, & Comber, 2018) and sustainable (Bødker, Korsgaard, & Saad-Sulonen, 2016; Ganglbauer, Fitzpatrick, & Comber, 2013; Heitlinger, Bryan-Kinns, & Comber, 2018) food systems, where sharing and redistributing surplus food are paramount. Much research has explored how digital technologies can play an important role in food waste reduction through a rising number of mobile apps specifically designed to reduce domestic waste by improving households' food management efficiency (Altarriba, Lanzani, Torralba, & Funk, 2017; Comber, Hoonhout, van Halteren, Moynihan, & Olivier, 2013; Farr-Wharton, Choi, & Foth, 2014). However CSCW and HCI scholars also have a burgeoning interest in more collective approaches for reducing waste, namely the many urban food-sharing initiatives that are popping-up across a range of diversified cities (Berns & Rossitto, 2019; Burton, Meier, Olarte, Skeini, & Zahan, 2017; Chies, 2017;

Ciaghi & Villafiorita, 2016; A. Davies, 2019; Ganglbauer, Fitzpatrick, & Molzer, 2012). Food-sharing initiatives have been proliferating as an activist response to the consequences of capitalist and global food chains (Schanes & Stagl, 2019). Food-sharing initiatives can take a variety of forms with some examples being web-based food networks (Prost, 2019), underground restaurants (Rombach & Bitsch, 2015), public refrigerators (Kera & Sulaiman, 2014), or within households consisting of non-related people for example, students (Falcone & Imbert, 2017).

Digital technologies have transformed the way we conceive of and (re)-distribute surplus food across different contexts (A. Davies et al., 2017; A.R. Davies & Legg, 2018), and research has outlined sociotechnical configurations of food-sharing through three emerging models, namely, for profit, for charity, and for community (Michelini, Principato, & Iasevoli, 2018). The work of FoodSharing Stockholm best fits the community model as it is a grassroots, volunteer-led community where surplus food is shared for free, without distinction between those who are accessing the services based on financial disadvantage or those who are simply making an ethical choice to prevent waste. Information communication tools have been key drivers in making sharing for community initiatives possible. Whether it is a specific online platform enabling consumers to reach each other (Ganglbauer et al., 2014) or making use of mainstream social media platforms such as Facebook (Berns & Rossitto, 2019), or messaging services such as Telegram (Engelbutzeder et al., 2020). Such tools can support the proliferation of food-sharing initiatives wherein one initiative provides a template for digitally infrastructuring another, thus reducing the efforts required to start up. For instance, the German foodsharing platform foodsharing.de began in Berlin but now facilitates multiple food-sharing initiatives across several German, Austrian and Swiss cites by providing digital infrastructure to facilitate surplus food collections and exchanges (Ganglbauer et al., 2014; Rombach & Bitsch, 2015).

A lack of social relationships and, consequently, of trust have been shown to have the most negative impacts on food sharing practices (Farr-Wharton et al., 2014; Lazell, 2016), researchers have explored the various ways sharing relationships between food-donors, volunteer organisations and food recipients can be configured and how they may be affected by digital technology use (Berns et al., 2021a; Ciaghi & Villafiorita, 2016; Ganglbauer et al., 2014). Previous work has identified the practice of food-sharing as a form of *community economy* (Berns et al., 2021a), that is, a generative economic system in which the process of negotiating interdependence is central (Gibson-Graham, Healy, & Cameron, 2016). Here, the authors describe how within a community economy, multiple layers of participation, commitment and interdependence are embraced. Moreover, it is recognised that degrees of involvement can and will vary among community members and change over time (Berns et al., 2021a). Past research has investigated the tensions associated within a peer-to-peer sharing in a broader sense, such as reciprocity, intimacy, and inclusion

(Lampinen, 2021), as well as those more specific to food-sharing such as othering (Ganglbauer et al., 2014), communicating activism (Berns et al., 2021a), and managing the flow on attendees (Berns, Rossitto, & Tholander, 2021b).

2.2 Organising Collective Actions

Past CSCW work has engaged with the collaborative efforts embedded in various types of activism such as how it is organised (Menendez-Blanco, Bjorn, & De Angeli, 2017; Mosconi et al., 2017), the role of sociotechnical systems such as social media (Erete, 2015; Ganglbauer et al., 2014), and the complex network of systems, information, people, values, narratives and ideologies that constitutes such activism (Bødker, Korsgaard, & Saad-Sulonen, 2016; Dickinson, Díaz, Le Dantec, & Erete, 2019; Wilson, Zhou, & Starbird, 2018). More recently, a JCSCW special issue has zoomed in on how activism materialises and evolves online. Resonating with earlier work from social movement studies (Klandermans & Oegema, 1987; Martin, Upham, & Budd, 2015), Hansson et al., (Hansson, Pargman, & Bardzell, 2021) unpack four dimensions that intersect and initiate activism online; (1) motivation meaning participants have a strong personal affect for a political issue, (2) culture that can be captured or reinforced through communication technology, for example, language, images, and interaction patterns, which frame the activism and situate it in wider narratives, (3) language and symbols that can link local activism with global movements; and (4) infrastructure i.e., information technology that sets protocols and rules for communication, provides the relations of production, and interacts to a great extent with other dimensions (Hansson, Pargman, & Bardzell, 2021). This connects to earlier CSCW and HCI work on publics (Dantec, 2016; Dewey, 2012; DiSalvo, 2009) that investigates the way design can draw people together on shared issues and provides an overarching framework to talk about infrastructuring community initiatives through design. This research has outlined the role of technology in providing social and material resources, creating attachments to shared matters of concerns, and sustaining people's engagement with and around them.

Work from digital civics also argues that successful civic interventions focus on innovative use cases rather than innovative technologies and that technology should be thought of as social and community infrastructure rather than a solution in itself (Schrock, 2018). Infrastructures are described as "reconfigurable" (Dantec, 2016) or "hackable" (Schrock, 2018) in that they are can adapt to and manage future changes in issues and attachments that a community may face. The use of Facebook across many grassroots, activist communities, is a prime example of this (see for instance (Ganglbauer et al., 2014) and (Rossitto, Lampinen, Light, Diogo, & Bernat, 2021).

CSCW research has highlighted hybrid forms of community engagement that are enacted through a combination of online and face-to-face interactions (Mosconi et al., 2017). The paper provides a concrete example of the local opportunities afforded by popular social networking technologies by unpacking how local citizens in Trento, Italy appropriated Facebook's group feature

to increase re resources that individuals and organisations in the community were able to mobilise offline (Mosconi et al., 2017). This connects to earlier work on artifact ecologies that has focused on how communities manage and organise a variety of digital technologies, to infrastructure their work, Bødker and colleagues (Bødker, Korsgaard, & Saad-Sulonen, 2016) have identified different phases volunteer-run communities go through: becoming a community, everyday community, and building anew. The first phase relates to the initial stages, when a community is formed and constituted. The second one betokens the work of sustaining collectives, while the last indicates how communities might change as they expand, and become connected to other actors. Overall, this work shows that such communities operate with limited resources, that their ways of working might change, and that they are often dependent on how and when volunteers can participate.

Moreover, specifically focusing on issues of scaling up (i.e., expanding what already exists), previous work (Biørn-Hansen & Håkansson, 2018) has illustrated how grassroots and community-driven initiatives are organised through different stages of development. The authors identify sustaining, growing, and spreading as key moments of these processes. Sustaining concerns activities aiming at bringing together the resources that communities need to operate and exist. Growing connects to communities' possibilities to operate beyond the initial member groups. Finally, spreading relates to communities attempting to define their impact by, for instance, disseminating skills, and knowledge. Resonating with Bødker and colleagues (Bødker, Korsgaard, Lyle, & Saad-Sulonen, 2016), we note that throughout these different stages, concerns of organising might be addressed as a consequence of contingencies and emerging tactics, rather than (pre)defined, goal-oriented strategies (Bødker, Korsgaard, Lyle, & Saad-Sulonen, 2016). Here, happenstance, that is the circumstantial presence of people, reflects on practical work, decision-making, and on defining key aspects of technology use, and organising more broadly.

Organisers and participants of social movements pursue social transformation motivated by a strong pathos for an issue rather than any particular political affiliation (Butzlaff & Deflorian, 2021). Prominent examples of online and hybrid in recent years have focused on specific issues such as the fight against sexism (Hansson, Sveningsson, & Ganetz, 2021; Rossitto et al., 2020), racism (Mundt et al., 2018), or climate change (Brünker et al., 2019) The efforts of these groups lie in changing society directly by altering everyday routines and offering alternatives to contemporary practices (Butzlaff & Deflorian, 2021). The infrastructural arrangements supporting such collective actions serve to bypass formal organisational practices while also lowering the cost of political participation (Dahlberg-Grundberg, 2016; Earl & Kimport, 2011). With this in mind, recent scholarship (Lampinen et al., 2022) has outlined how initiatives might get started, or evolve in unexpected ways, as the consequence of something that travels or that is moved forward to other initiatives. This work has proposed the use of "proliferation" as an alternative to the notions of replication and growth that generally frame mainstream discourses

on computing. The authors focus on processes of proliferation to illustrate how community initiatives are created and evolve as artefacts, practices, or social ties find their ways in new contexts and places. Connecting with this, other research (Rossitto, 2021) has illustrated how initiatives might be interconnected as meaningful outcomes and experiences are circulated, and digital templates for participation shared.

2.2.1 Grassroots Governance

Tightly connected to issues of organising collective action, is governance, particularly grassroots governance. Grassroots communities provide significant examples of how people self-govern collective action to address diverse concerns. Past work from HCI and CSCW has investigated the intricacies of self-governance across different contexts such as maker-spaces Schmid (2021), community-based recycling organisations A.R. Davies (2009), feminist activism Hansson, Sveningsson, and Ganetz (2021); Rossitto et al. (2020), online communities (Dosono & Semaan, 2019; Schneider, 2021a), open source software development communities (Schneider, 2021b), and platform co-operatives (Cherry, 2016). These examples draw attention to the myriad of challenges that can arise while striving for grassroots democracy, such as enabling flexible participation, juggling burn out from volunteer work, documenting work and transparency, negotiating rules and norms, balancing structure and openness, managing emotional labour, and balancing community ethics, ideologies and pragmatism. In this paper, we align with scholarship that describes grassroots democracy as being *"practised through a system of norms, values, societal processes and institutional arrangements fueled by the commitment and capacities of ordinary people"* (Tandon, 1997, p. 4), rather than as a system based on formally defined rules and procedures. This point is echoed by Dana and colleagues who in their book, 'Community Rules', (Dana, Hornbein, Russell, & Schneider, 2021) stress that rules can be important and needed but can also constrain actions and opportunities to act, develop and adjust.

However, research shows that grassroots communities do benefit from having mechanisms for democratic decision-making (Dana et al., 2021) and past work shows how finding a balance between formal and informal governance structures can be difficult. For instance, technology and democracy scholar Nathan Schneider (Schneider, 2021a) exemplifies how despite the democratising potential of the Internet, technology often assumes that there should be admins or mods with nearly absolute power to censor or exile members when conflict occurs. While in another article, the same author (Schneider, 2021b) warns of the 'tyranny of openness' that he has seen emerge in peer-production communities with regards to open source software licensing. The latter publication is inspired by Jo Freeman's 1972 essay "The Tyranny of Structurelessness." (Freeman, 1972). The central argument of this essay is that structure will exist whether formalised or not, and that in groups that purport to be leaderless and non-hierarchical will simply end up with leaders and

hierarchies emerging anyway; this can happen among socially privileged participants or more active volunteers ultimately creating communities he regards as undemocratic. To counteract this Freeman encourages activists to adopt simple, explicit forms of "democratic structuring" and make power structures that participants can use or challenge visible (Dana et al., 2021; Freeman, 1972)

3 FoodSharing Stockholm

FoodSharing Stockholm is a volunteer-run, grassroots community that aims to (re)-distribute surplus food, donated by restaurants or big-chain supermarkets, that would otherwise go to waste. The community was established in September 2019, and it is made up of five participant groups who are brought together by concerns for food waste reduction. *Food-sharers* are the volunteers who run the community; they set up co-operations with food-donors, collect, sort and share surplus food as well as rescuing and eating it themselves.*Food-savers* are individuals who rescue food from going to waste by attending events or by stopping by the community's solidarity fridge to collect surplus food. *Food-donors* are the businesses such as grocery stores, restaurants and cafes that donate surplus food. *Hosts* are the establishments/locations in which food-sharing events are held. And finally, the *Collaborators* are any other charitable, activist, or volunteer-led organisations with whom the community collaborates by supplying surplus food. The use of these terms was negotiated early on by the core volunteer group to establish a clear language for describing the various participants. Members of the community were searching for terms beyond the standard approach of "volunteer" and "participant" in an attempt to cultivate a culture of collaboration in the community, where all members save food together, in opposition to the more typical binary narrative of active and passive participant (McCarthy & Wright, 2015). In the paper, we refer to them by adopting the same names that early members of the community have decided to use. We, however, recognise that the use of these terms can also be problematic; for example donors are also sharers, and volunteers are savers, arguably even more than participants, since they put in a lot of work to make events run.

At the time of writing the community has been active for over three years, and participation has fluctuated. The food-sharer (volunteer) and food-saver (recipient) groups have been those that have been subject to this fluctuation. For example, the first meeting in September 2019 had twenty-four people in attendance, in September 2020 there were approximately eighteen active food-sharers (volunteers), and in September 2021 eight active food-sharers (volunteers). Likewise, the community has experienced periods of high and low turn out to food-sharing events with some events attracting over thirty participants and others a little as four participants. Hosts have increased gradually over time as food-sharing events scaled from one to three. Community partnerships with food-donors have remained quite consistent with one supermarket and one anti-waste focused restaurant (that sells meals cooked from surplus

food) supplying the majority of the food shared at events since the beginning. Collaborations with other organisations have also happened, although on a more sporadic basis; for instance, the community has supplied food to the local chapter of the global environmental movement Extinction Rebellion [2] for demonstration events, as well as, supplying specific food items to a nearby community cafe that offer a free breakfast to guests in need each today. At present, FoodSharing Stockholm is sustained with twelve active food-sharers (volunteers) who organise three weekly surplus-food distribution events, hosted at three different locations, and attracting between 15 and 25 participants. Together the community have rescued over nineteen-thousand kilograms of food.

3.1 Data Collection

The engagement with the community has been framed by a Participatory Action Research (PAR) approach (Hayes, 2011, 2014; Kindon, Pain, & Kesby, 2007), with the first author being involved as both a researcher and active food-sharer (volunteer) since it was founded in 2019. Methodologically, the research project combines contextual inquiries and first-hand understanding of the setting with a practical orientation towards the community's work. This means that besides understanding the processes around the community organisation, the study's aspiration has been to support the community practically, for instance, by taking over responsibilities, and providing guidance on the adoption of existing digital technologies, and the design of sociotechnical practices.

Having seen the call to participate in the group on a local Facebook page, the first author commenced a long-term ethnographic study of the newly formed food-sharing community. At the time of writing, engagement with the community has lasted for three years; the data presented in this paper have been collected between September 2019 and September 2022, mostly by means of complete participant observation (see (DeWalt & DeWalt, 2002)) carried out at meetings, sharing events, and during the processes of enrolling new food-donors. Complete participant observations entail the researcher's active role as both an observer and a core member of a social group. As the community is in the phase of maintaining main practices and collaborations with third parties, the research engagement with it has been valuable to understand aspects such as sociotechnical practices of infrastructuring the community's efforts, making them sustainable over time along with the food-sharers' (volunteers) attitudes towards growth.

These data consist of: (1) 66 single-spaced typed A4 pages of notes and minutes from thirty-nine volunteer meetings; (2) 43 single-spaced A4 pages of weekly reflections and field-notes documenting activities that took place during food collections and food-sharing events, and conversations between community members that took place in face-to-face and online settings. The

[2] https://rebellion.global/

quotations found in the analysis derive from volunteer meetings and the conversations. These data provide rich accounts of both the minutiae of the day-to-day work of setting up and maintaining a surplus food project while also documenting higher level discussions around complexities of community organisation.

3.2 Data Analysis

The data analysis was carried out on a continuous basis throughout the study, with the first author writing reflective notes and highlighting working themes after each meeting. This meant, for instance, mapping out the meta level aspects of the community that were visible to the first author through repeated participation. In the first round of mapping, three central aspects of the community were identified: people, tools, and surplus food. Starting with "people" a visual map was created illustrating who the actors of the community are, such as the food-sharers (volunteers), food-savers (recipients), food-donors, hosts and other collaborators that make the rescue and redistribution of food possible. Once this map was created, an accompanying text was composed to describe the role of each participant group, and to situate it within the context of the community, and relations with other actors. For example, closely examining the "food-sharer" group included observations on how they are the volunteers do the groundwork for the community, their past experiences, and the diverse cultural perspectives they may bring to the community. The same process was followed for the subsequent "tool" and "surplus food" categories– first creating visual maps, then building on it with a descriptive text. The "tool" map included central digital tools such as the community's website and Facebook page; each tool was then connected with the actors that interact with it, and a short text describing what they used it for – e.g., the Facebook page is used for communication between food-sharers (volunteers) and food-savers (recipients). The "surplus food" map visualised the flow of the food stating, from collection from the food-donors, showing what food items are typically collected and where and how they are eventually shared with food-savers (recipients) in different ways. These maps worked as an analytical framework to connect codes and themes to a broader picture of what the key elements of the community

As the data collection progressed, the first author gradually triangulated this initial map with thematic analysis (Braun & Clarke, 2012) of the subsequent field notes and weekly reflections that documented their participation. This process began by highlighting words, phrases and sentences in the text, and identifying shorthand labels or "codes" to describe their content. These codes captured a more detailed account of the day-to–day operations of the community as well as her own personal reflections as both a participating volunteer and researcher. The data became richer and more nuanced over time, for instance, the initial category of "people" expanded to include codes such as "community not service" and "conflicts/disagreements between food-sharers (volunteers)", the initial category of "tools" expanded to include codes such

as "building routines" and "online/offline balance" and the category of "surplus food" include codes such as "fairness" and "managing expectations". In final rounds of analysis the authors worked together to iteratively unpack the various connections and relationships that exist between these categories, and codes were sorted into the following themes: setting-up the community, setting-up sharing, and governing the community.

3.3 Reflections on ethics and positionality

The presented research started as a result of the first author's wider interest in food-sharing initiatives. Her research involvement with another food-sharing community, FoodSharing Copenhagen (Berns & Rossitto, 2019; Berns et al., 2021a, 2021b) provided insights on what would be critical issues to practically deal with (e.g. enrolling volunteers, organising food pick-ups), and interesting analytical aspects to work with (e.g., the challenges connected to running sharing events). Building on this past experience, the adoption of a Participatory Action Research approach (P.A.R), was central to not just studying, but also gaining first-hand experiencing of the work of establishing, sustaining and growing a food-sharing community from the ground up.

As with our previous work on (Berns & Rossitto, 2019; Berns et al., 2021a, 2021b), the choice to not anonymise the name of the food-sharing community was negotiated with the community members. This decision resonates with HCI and CSCW research that has called for a reconsideration of ethics in anonymisation practices (Brown, Weilenmann, McMillan, & Lampinen, 2016), and for a concern to give credit to community-led initiatives striving for social change (Rossitto & Lampinen, 2018). This approach also reinvigorates the argument that HCI research should find sites of resistance, narrate them, and help them proliferate through design research and practice (Heitlinger, Bryan-Kinns, & Comber, 2019). The geographical location of the community has been disclosed, as it is important to recognise the very specific socio-cultural context that shapes the ways the community operates and conceives of surplus food. However, the names of the individuals who participate in the community and this study have been anonymized.

Maintaining and juggling both a researcher and a participant role has often been a conscious balancing act. Throughout the research process, the first author made explicit choices as a researcher and others more as a member. For instance, through previous involvement with research and other communities, they were aware of some challenges that would likely arise while setting up such an initiative (e.g., defining a vision), and such knowledge was shared with the rest of the community members, thus shaping the way problems were framed and addressed. On several occasions, the first author decided to prioritise their role as a participant to engage in "participation with the other rather than for professional privilege over them" (McCarthy & Wright, 2015, p. 15). In practical terms, this sometimes entailed forgoing more data collection, such as conducting planed on site interviews or workshops, to support the running of food collections or sharing events because there was a last minute shortage

of food-sharers (volunteers). However, this approach of complete participant participation allowed for the relevance and trustworthiness of the data analysis to be explored continuously by means of member checking (Baumer, Xu, Chu, Guha, & Gay, 2017). As an active member of the community, the first author could check their interpretations of the data against the perspectives and understandings of fellow members openly and on an ongoing basis. Moreover, our contribution on analytical insights and practical sensitivities reflect the first author's dual roles of being both a researcher and and a community member.

McCarthy and Wright (McCarthy & Wright, 2015, p. 15) discuss how there is a tendency in research to keep "a professional distance" from the subject of inquiry to be able to document and learn about it. They describe this as "an emotional distance that enables the researcher to produce an organised account by playing down the disorienting messiness of everyday experience" (McCarthy & Wright, 2015, p. 15). During data collection and analysis, the first author has often experienced same messiness, which was also described by their fellow volunteers. For example, they felt the anxiety that accompanies coordinating events, and endured the emotional labour of feeling unequipped to adequately support members of the community who were experiencing food insecurity. This messiness is reflected in the points raised in the discussion, for instance, how organising activism around surplus food is circumstantial and highly variable, or how decision-making models require constant renegotiation. To counterbalance this feeling of "messiness" and to maintain a level of reflexivity throughout the study, the first author has kept a diary where they documented personal feelings, biases and insights experienced during sharing events and community meetings.

In a longitudinal study such as this, with participants and volunteers coming and going, ethical concerns such as obtaining informed consent from all participants can be problematic. To overcome related challenges, we took a two-pronged approach. Firstly, verbal consent to participate in the study was collected on an ongoing basis with participants as the first author spoke openly about the research project and what it means to 'participate'. Secondly, during the three month time-period between data collection and submitting the paper for publication, the first author circulated a digital consent form outlining the study results and that all personal data would be anonymous, but the name of the organisation would be revealed. This consent form was used by the authors to obtain explicit consent from all community members whose direct quotes have been used.

4 Analysis

The analysis is structured around three themes that capture how the articulation work of setting up the FoodSharing Stockholm community was carried out by drawing on the success stories, strategies, and key practices of other similar communities. The first theme examines the central sociotechnical activities

of setting up the community, namely, developing ways of working, recruiting food-sharers (volunteers), and establishing partnerships with food-donors. The second theme focuses on the negotiation of values and practices framing the re-distribution of *surplus* food, unpacking the mutual influence of both the activist concerns and logistical practicalities con of sharing, as well as negotiations of fairness within the community. Finally, the third theme deals with how governance and decision-making models are (recurrently) negotiated, and how the work of activism is maintained over time.

4.1 The Role of Previous Experience in Setting-up the Community

Dealing with the practical concerns to get the community started and make it work was the primary focus of early volunteers' meetings. In what follows, we detail how both practical arrangements, concerned with putting together the resources needed to operate, and efforts to define driving visions and values were shaped by participants' direct or indirect experience of other food sharing initiatives.

4.1.1 Developing ways of working

Showing how initiatives striving for social change often develop within broader contexts where specific matters of concerns have gained attention, FoodSharing Stockholm stems from the early actions of four people who shared concerns for environmental sustainability. While they did not know each other very well, these persons had discovered a shared interest in food-waste reduction when meeting at an environmental protest, which thus provided a relevant backdrop to start discussing their experience and views on the topic. This unplanned encounter was the starting point for a number of discussions and efforts that led to a kick-off meeting to which potentially interested people were openly invited. The invitation to join the initiative was shared openly by word of mouth within the personal networks of the initiators, and through a flyer that was distributed physically, by email, and various social media platforms. With twenty-four participants in attendance, the interest in what can be regarded as the community's first meeting was high.

The initial work centred on outlining ideas of how this food-sharing community could operate, and what visions about food sustainability could frame its operation. Early participants were mostly expats who had relocated to Stockholm and had previous experiences with similar initiatives in other cities and countries. As it stems from conversations between the present participants, their previous first-hand understandings of the challenges of organising food sharing were key to having focused, early discussions on framing underlying motivations for distributing surplus food. Sharing success stories from other food-sharing communities in was, therefore, instrumental to initial planning – for example, FoodSharing Copenhagen in Denmark, Solikyl (translation: Solidarity Fridge) in Göteborg Sweden, Fairteiler (translation: Fair Dividers) in

Vienna Austria, and foodsharing.de operating in various locations across Germany. As a result of this, the initial stages of setting up FoodSharing Stockholm was characterised by careful considerations of the ecology of framings (e.g., for charity, environmentally concerned), practices (e.g., how to structure distributions), and values associated with different experiences of food-sharing, both as a concept and practical ways of organising community work. Food-sharers identified how some groups viewed food-sharing as a form of solidarity (Chies, 2017) while others appeared to be more driven by environmental concerns (Berns et al., 2021a) associated with wasting food.

In this early phase, knowledge of other initiatives was also useful to reflect on how available human resources (i.e., number of volunteers, food donors) would relate to the scale of operation to aim for. Here, it was outlined that some communities had grown to be quite large organisations and food-sharers, who had previously participated, shared their opinions about them. For example, it was explained that FoodSharing Copenhagen was a highly structured organisation with an active board and a formalised structure for recruiting as well as training food-sharers (See (foodsharingcph.org, 2020) for details). Similarly, foodsharing.de was also described as being a structured organisation, and food-sharers who had previously participated in it credited their bespoke platform (foodsharing.de, 2020) for its success to scale out its activities across different locations. Although inspirational, the large-scale operations of these were experienced as intimidating. Participants, were aware that this level of organising would be difficult to replicate with an unknown number of food-sharers and food-donors. Therefore, food-sharers found themselves drawing more on the experiences of smaller and more localised communities such as Solikyl (solikyl.se, 2019) and Fairteiler (fairteiler, 2022). Staying small, to operate with limited capacity in terms of human assets, was an explicit strategy agreed on already in the phase of setting up the community.

A third issue extensively discussed was the work that is practically needed to make the community work, and how related tasks and responsibilities could be distributed and organised. These tasks were largely identified through a brainstorming session. Here as well, the participants' direct experience and knowledge of other food communities played a central role in outlining the main concerns to be dealt with. Early on, three working groups were established: 1) a communication group, that would work on strategies for finding food-donors, finding locations for sharing events, for advertising to potential food-savers (final recipients), and developing a mission statement for the community's main motivations and the goals it strives for; 2) the fair-share group, who planned to investigate synchronous and asynchronous modalities for sharing food fairly (e.g., hosting events, setting up a community fridge or pantry); and finally, 3) the technology group was responsible for researching digital tools that could be useful for organising and structuring different aspects of the community's work.

The explicit distribution of responsibilities to each of these groups was a strategy to divide the processes and work of setting up the community

into specific issues. This strategy was, however, difficult to uphold. While the early participants' previous experience with other food-sharing initiatives had helped identify key operational needs, it did not extend to the practical orchestration of them, especially when operating with limited capacities. Shortly after the working groups was formed, the total number of participants reduced from twenty-four to twelve, after a number of initially interested people dropped out early on. The reduction in numbers resulted in the decision to hold all subsequent meetings as one group. This was the first of many instances where food-sharers needed to adjust and re-organise the community's work based on changing circumstances (e.g., a food donor changing the food collection day) or changes in participation numbers (this is detailed further in 4.2).

4.1.2 Creating a digital infrastructure

As previously mentioned, researching digital tools to use for organising and structuring the different aspects of the community's work was a key focus for setting up the community. Significant efforts were put into strategically planning the different digital tools that could form the community's digital infrastructure. The first step towards this took place at the first meeting where the community initiators introduced a platform called *Karrot* (karrot.world, 2022). It was explained that they had learned of the platform during a discussion with a member of the neighbouring Solikyl community. Karrot was presented to other attendees of the first meeting as a free and open-source tool for food-sharing groups that allows initiatives to create independent 'groups' with their own name, logo, processes, rules and agreements. Most attendees agreed that it sounded intriguing, but some (including the first author) voiced concerns about having the majority of interactions take place online rather than face-to-face. Therefore, the platform was not adopted straight away, but instead it was agreed that it would be investigated as a potential tool by the technology working group. Over the course of a month, a FoodSharing Stockholm group was created on Karrot to enable food-sharers to explore and experimented with the features offered by Karrot, such as creating events and making the location of potential donors on a map. Following this experimentation period, Karrot was ultimately adopted, and remains as the central infrastructure used by food-sharers to organise face-to-face activities food collections, distributions and meetings as well as to collect statistical data on the quantities of food distributed. Telephone contact and SMS messaging was also initially used between food-sharers if quick contact was required as not everybody had the possibility to check Karrot regularly as a mobile version of the platform is available only on Android and not available for IOS users. However, in early 2022 SMS messaging was eventually swapped out for a WhatsApp group to facilitate group discussions transparent to all food-sharers, rather than one-to-one conversations.

Resonating with previous work on artefact ecologies (Bødker, Korsgaard, & Saad-Sulonen, 2016), the first two years of operation of this community shows that the digital infrastructure needed to be flexible, and that the adoption

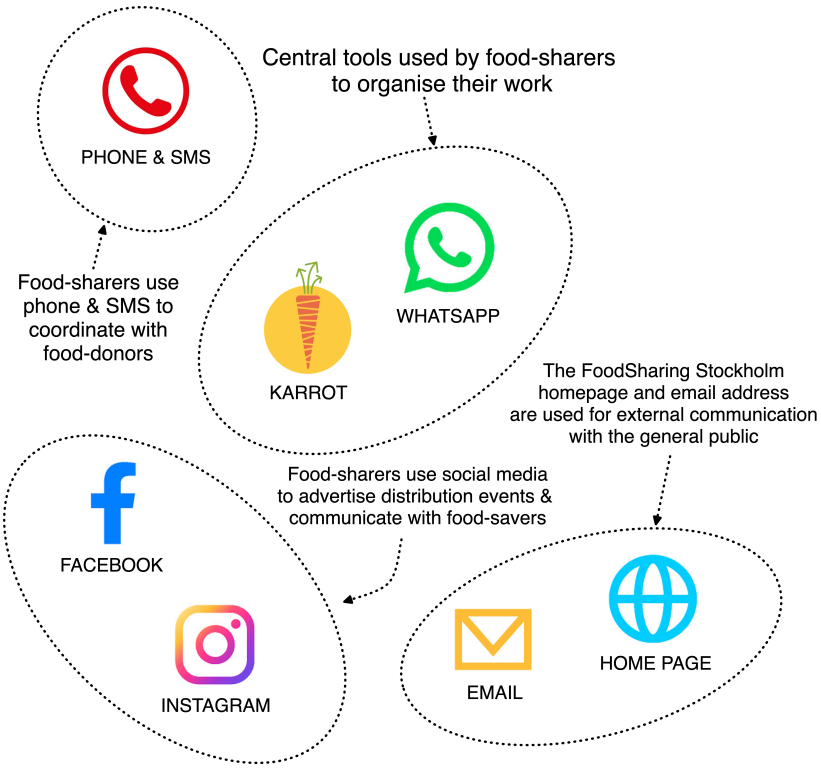


Fig. 1 This overview of the digital infrastructure of FoodSharing Stockholm illustrates the technologies that the community use and their primary functions.

of specific technologies, or functionalities within them, also evolves from situated circumstances. The digital infrastructure of the community continued to expand as different needs arose. For many food-sharers, it was important to use open source, non-mainstream tools, such as Karrot as much as possible, as they felt such tools would better align with the values of the community. Most tools required, such as email and web hosting, were easy to find as open source. Yet, to gain local visibility and attract people to attend food distributions, food-sharers felt that it was also necessary to have a presence on mainstream social media platforms, such as Facebook and Instagram. A practice that had proved successful for other food-sharing communities, such as FoodSharing Copenhagen and Solikyl. As a result of using Facebook and Instagram for advertising food-sharing events, food-savers began to reach out using the same platforms, resulting in Facebook and Instagram direct messenger becoming the primary means of communication between food-sharers and food-savers as well as the most common first point of contact for potential new food-sharers. The decision to use the mainstream social media platforms connects to other work (for

example, (Rossitto, Lampinen, et al., 2021)) that highlights how the convenience connected to adopting available technology might override concerns for digital platforms' economic interests or (miss)alignment with a community's values.

4.1.3 Enrolling food-sharers

As also noted in other studies (Biørn-Hansen & Håkansson, 2018), enrolling food-sharers (volunteers) and creating the social assets needed for the everyday work was central to setting up the community. Here, relations to other food-sharing initiatives were instrumental to reflect on enrolment practices that were maybe not best suited to current circumstances, as well as outlining possible problems related to variable degrees of commitment to the community's values.

In the case of FoodSharing Stockholm, an initial strategy adopted to facilitate the enrolment of food-sharers was to individually respond to all private messages, sent on Facebook or Instagram, by people who showed an interest in the initiative. This was, however, experienced as too laborious, to the point that food-sharers felt it was taking time away from the needed, practical work of collecting and distributing food. As it was not uncommon for 'interested' persons to quickly lose interest or never actually participate, the responsible food-sharers grew frustrated that time invested in this tasks seldom resulted in increased participation. To rethink the approach, the core food-sharers decided to look at how recruitment was done in other food-sharing communities. Here, both instances of happenstance and more structured enrolment practices were considered. For instance, some food-sharers recalled their experiences of becoming involved by chance, after stumbling upon a sharing event or seeing an event advertisement on social media. Relatedly, food-sharers who had previously been involved with larger collectives such as FoodSharing Copenhagen (foodsharingcph.org, 2020) provided examples of more formal online on-boarding systems, where potential new volunteers would get to watch informational videos about the organisation and about what being a volunteer entails. Followed by a short quiz intended to determine if potential volunteer understood what Foodsharing Copenhagen is all about.

As it emerged from related discussions, food-sharers found this process difficult and time-consuming to implement, as well as too formal and impersonal. Instead, to stay flexible, food-sharers decided to set up multiple digital and physical contact points that could automate part of the process and filter general enquiries from more constant involvement. With this sociotechnical setup, the first contact was an automatic reply on Facebook Messenger that thanked senders for reaching out, and linked them to a website with a description of the community, information on how to get involved, as well as an FAQ. On the website, anyone interested in becoming a food-sharer was invited to stop by at one of the distribution events, to join one of the biweekly meetings, and then join the core working group on Karrot. At this point, one experienced food-sharer would take on the responsibility to welcome newcomers personally, guide, and assist them during their first shift. Apart from being more flexible,

this system also limited access to the Karrot platform and, therefore, access to sensitive contact information about partnering food-donors. Again, based on direct experience of other food-sharing communities, some of the volunteers were aware that such information had been abused, with people trying to receive donations outside of the prearranged times, resulting in breakdown of the collaboration.

4.1.4 Establishing partnerships with food-donors

Developing strategies for approaching potential food-donors, and avoiding potential pitfalls was unsurprisingly one of the most prominent practical concerns in setting up the community. Food-sharers drew on the challenges faced during past experiences with similar communities, as well as, learning from strategies for recruiting food-donors that were published in a online anti-food waste forum, foodsaving.world (foodsaving.world, 2022)

Expanding on previous studies (Biørn-Hansen & Håkansson, 2018), in this section, we disentangle how this process intertwines with the food-sharers' effort to develop a clear vision for the community as well as with the development of a strategy to talk about food waste with potential donors. Much was learned from food-sharers' experiences with other food-sharing communities. For instance, past participants in foodsharing.de had experienced that a challenge with enrolling donors is that one only has one chance to convince them to collaborate with your organisation, in that store managers are busy and meetings are difficult to schedule. With this in mind, food-sharers agreed on the importance of being well prepared with clear information on what was expected from donors, and how food would be handled and shared. While planning how to approach potential donors, it became obvious that contacting donors required the communication of an explicit vision of the community and its goals (e.g., helping others in need, addressing sustainability issues), and a clarification of the potential benefits for the donor.

Implementing its own effective communication form, volunteers who regularly attended the community meetings, wrote a script to be used as a base to talk to donors. It was articulated in three steps to: a) enquire about potential food-donors' current practices for managing surplus/reducing food waste; b) explain what the FoodSharing Stockholm community seeks to achieve and how; and c) highlight the benefits a collaboration could provide. Two information artefacts were developed to support the food-sharers with this activity, namely a Q&A list, and a pamphlet with information about the community. The script was to be used as a guide to introduce the organisation in a clear and consistent manner. It consisted of a list of questions that donors could potentially ask, and responses to them. In the transcript below "Q" are the potential questions asked by donors, and "A" the suggested answers for volunteers:

Q. *"Who is food shared with and why?"*
A. *"Food is shared with everyone and anyone, the goal is to to show*

people that surplus food is not 'waste' and can be accessible to everyone"

Drawing on the past work of a similar community, other questions were adapted from a detailed guide for "How to build and maintain co-operations with stores" shared by another organisation on a community forum [3] on the Karrot platform. This included topics such as: (1) which store to approach, (2) how to initiate contact with store managers, (3) ideas on how to navigate common roadblocks, such as managers who like the idea of food-sharing but make claims such as, they don't throw anything away, and (4) what to do once a co-operation is attained, how to maintain it through reliability and professionalism. For example:

Q. *"We are already collaborating with another organisation, so why should I collaborate with you too?"*
A. *"If you are able to donate all of your surplus food to this organisation, great! Our aim is to avoid and minimise food waste so good job, we congratulate your efforts. If however you do have some food left still, we can help to reduce your surplus even further."*

To complement the script, pamphlets were prepared with more detailed information that would allow potential donors could review later on.

When approached, many businesses responded positively showing that they shared the concerns of the community, yet declined the invitation to join for a number of different reasons – e.g., ongoing collaborations with charitable organisations, previous problems with similar initiatives to donate food, concerns for the work needed to collect and work out surplus food, regulations about food safety, claims about not having any food waste. Although food-sharers tried to be strategic, putting considerable effort into preparing for meetings with potential food-donors, much of the work of enrolling food-donors played out as in-the-moment responses to the concrete, unexpected challenges of talking to them. Ultimately, food-sharers realised their most successful way to form co-operations was through utilising personal connections, with the hope of building enough trust with donors to expand to a more regular agreement. For instance, the first food donation, received by the community, came about through a friend of the first author who was employed in a supermarket, and who arranged a meeting with the manager to discuss a donation for one specific event at a local community centre. This cooperation later developed into regular, twice-per-week donations which are still maintained at the time of writing. Telephone calls and SMS messaging have become the typical means of communication with food-donors as communication is fast and does not require specific applications. As discussed during one food-sharer (volunteer) meeting, the combination of the non-committal 'one off' nature of the donation, coupled with trust deriving from the direct connection with the employee, has been regarded as a reason for this successful partnership.

[3] http://foodsaving.net/

4.2 The Role of Previous Experience in Setting-up Sharing

This section further illustrates the challenges of setting up the community (Section 4.1) by focusing on the negotiation of values and practices framing the distribution of surplus food. Alongside lengthy discussions about how to develop co-operations with food-donors, early on food-sharers also engaged in many conversations about the logistics of actually sharing surplus food, including motivations to share food with specific cohorts of people. Below, we illustrate how sharing surplus food is defined by visions and values, but also shaped by the practical, logistical challenges of collecting, sorting, and (re)distributing such food.

4.2.1 Considering and adopting existing food-sharing models

As noted in 4.1.1, volunteers decided, early on, that the primary goal of the community was to prevent food waste by sharing it with anyone. This approach foregrounds principles of solidarity and environmental sustainability, rather than the economic needs of more vulnerable groups. This is not to say that early food-sharers disregarded the role of charity. Many of them, with previous experience with other food-sharing communities, were just aware that charity initiatives require a more stable organisation and infrastructure to offer reliable support to those experiencing food insecurity. Thus, not being able to predict the flow and intake of donated food — especially with the small network of food-donors Food Sharing Stockholm partners with — also contributed to the vision of a food sharing community that would be primarily concerned with more sustainable food consumption.

Food-sharers (volunteers) drew considerable inspiration from other food-sharing communities to work out the logistics of sharing surplus food and adapt them to the context of a different city and community. Both synchronous and asynchronous models for food sharing were carefully examined. As noted earlier (section 4.2), during one early food-sharer meeting, four known models of food sharing used by similar communities were identified. One model, directly inspired by foodsharing.de (see (Ganglbauer et al., 2014)), involved one-to-one exchange between community volunteers and food recipients of pre-made surplus food boxes, following an online sign up process; another approach, inspired by the Solikyl community, was to have a community/solidarity fridge that volunteers would fill when donations were received, thus allowing food recipients to stop by at a time that was convenient for them (see (Chies, 2017)); another option, employed by the Fairteiler group, was a more direct form of sharing, where members simply took turns collecting from donors in larger groups with no distribution point (see (fairteiler, 2022)); finally, another option was an event-based model where larger quantities of food were collected, sorted by volunteers, and distributed to recipients/attendees at set dates. Two of these models, the community/solidarity fridge model, and the food-sharing event model were discussed in depth as potential options for FoodSharing

Stockholm. Both models were compared with regards to access to physical infrastructure (i.e., sourcing and placement of a fridge or finding a location for hosting events), time flexibility around food accesses and distribution, and potential impact (e.g., which model would enable the greatest amount of food to be shared, or raise people's awareness about food waste). Eventually it was decided to begin by adopting the community fridge model, in that it would provide much needed flexibility, and require little organisation work as the fridge can be filled anytime food is available.

Moving forward with the agreed strategy to share food asynchronously through a fridge, a partnership was established with a local, volunteer-run community centre where space was available. This connection was established through personal connections where a FoodSharing Stockholm food-sharer was acquainted with a member of the centre. However, quite early on this co-operation became problematic as the location did not have regular opening hours, thus making it difficult to accurately advertise opportunities to collect food. This led food-sharers to adjust to a hybrid approach, where most of the donated food was shared at planned, face-to-face events, with any remaining food being placed in the fridge for collection when the community centre was open. Combining these two ways of sharing gave the community a combination of visibility and flexibility. Bringing food-sharers and food-savers together for events was advantageous for rendering visible and developing activism around food waste as a societal issue. At the same time, having the fridge made food collection more flexible by expanding collection possibilities and taking pressure off food-sharers to distribute all food items by the end of each event. Communication with food-savers was predominantly carried out through social media, namely Facebook and Instagram, and mostly to advertise events. The two digital platforms were used to share information about distribution times and location. Images of food that has been saved, and meals that could be cooked with it were also often shared.

4.2.2 Negotiating fair sharing

Complementing the practicalities of how food is distributed are questions of what values should shape food sharing. Although, from the perspective of food-sharers, the central goal of FoodSharing Stockholm is to reduce food waste, they also feel responsible to try to share available food as fairly as possible among those who attend events. However, multiple discussions between food-sharers and between food-sharers and food-savers revealed that there was no shared agreement of what constituted 'fair' sharing within the community. The first author had a first-hand experience of this issue in her previous work with the FoodSharing Copenhagen community (Berns et al., 2021a, 2021b). Community members also felt that the most 'fair' but also efficient way would be to count all of the food items and divide them equally between the number of food-savers present that day. While others felt that this approach was not really conducive to reducing food waste as it might mean that people collect food that they may not actually use. This raises the point that 'equal' may

not always equate to 'fair' for example, perhaps it does not make sense for a person who is just feeding themselves to collect the same amount of food as say someone with a family of four. Similarly, some community members felt that personal situations should influence who gets to take what, for instance two older food-savers suggested that those who are old should get to collect food first.

Recognising that there is no clear solution to this, food-sharers, once again, looked to similar established communities for practical inspiration. Inspired by an approach previously used by FoodSharing Copenhagen, food-sharers implemented the following system at sharing events: As food-savers arrive they each blindly pick a numbered ticket out of a box, this ticket decides the order in which they will collect food, like a lottery. The numbers are then called in sequence and food-savers are asked to decide how much food to take based on what they will be able to consume while also trying to consider others who are also hoping to collect food. Food-sharers use their experience to help to guide food-savers on what a 'fair' amount to take might be and food-savers who wish to can partake in a second collection round, once everyone has gone once. This allows for flexibility on what to collect but also ensures that even the person with the last number in the queue has a chance to collect something, and that all of the food will be distributed. As one food-sharer stated in a discussion thread on Karrot:

> "I think that it's okay to come out of the distribution and one saver got 5 bananas and no one else got bananas but someone else got 5 apples and someone else got 5 oranges." (Food-sharer 4)

This approach to sharing surplus food resonates with the point made above that issues relating to environmental concerns, social responsibility and logistical challenges are interconnected in food-sharing communities. Members of FoodSharing Stockholm try to adapt to the complexities of food waste reduction and reinvest surplus food with new value. Events focus on reducing waste at retail level but also work to encourage all members of the community to be more aware/conscious of reducing waste at home. For instance, when welcoming food-sharers to the events food-savers make a speech describing how the event will work and offer the guideline that *"just because it's free doesn't mean you should take as much as possible, think about what you can really use"* (Food-sharer 5)

4.3 Exploring Governance Models

Processes and strategies to govern and organise the food-sharing community were negotiated and structured over time. Once again inspiration was drawn from food-sharers' direct experiences with other activist groups, within and beyond the context of food-sharing, as well as, direct conversations with members from the neighbouring community, Solikyl. The data highlights how efforts to define governance unfold resonating with scholarship that describes grassroots democracy as *"practised through a system of norms, values, societal*

processes and institutional arrangements fueled by the commitment and capacities of ordinary people" (Tandon, 1997, p. 4). In what follows, we detail the ongoing processes of negotiating different organisational and decision-making structures and the day-to-day processes that reinforce them; this includes, for instance, discussions between food-sharers (volunteers) on the topics of reaching consensus, scaling the initiative, and reflections on when it would be a suitable time to become a more formally structured organisation (e.g., an NGO).

4.3.1 A non-hierarchical model

In early discussions on how to run the community, one of the core food-sharers proposed adopting a non-hierarchical, democratic decision making model. This became the topic of several subsequent meetings with much time was spent discussing how a non-hierarchical model would work in practice and the pros and cons of the model – for example, most food-sharers agreed that power would be equally distributed, while also sharing concerns about possible lengthy deliberation processes. The organisational approaches of other similar groups were considered as relevant examples. For instance, some food-sharers discussed their experience with the highly structured approach of FoodSharing Copenhagen that included a biannually elected board and six working groups who managed various aspects of the community (Berns & Rossitto, 2019). Additionally, a video call was arranged with volunteers from the neighbouring community Solikyl as a way to learn from the experiences of another Swedish initiative. During this call one volunteer advised the community during a video meeting that having a hierarchy should not be overlooked. Echoing past work which argues that structure will exist whether formalised or not (Freeman, 1972; Schneider, 2021b), the Solikyl volunteer advised that hierarchies can be useful to solve problems as there will be someone who can make difficult decisions, and that a more horizontal model requires an explicit agreement on what can be decided by a small group of people, and what instead requires broader consensus:

> "Hierarchy can actually be good once people don't abuse power, [a hierarchical model] does support mechanisms to move out of difficult situations [...] but its important to consider, what are the freedoms within the group, and what sorts of things should be agreed as a group? Sometimes people need to be pushed to take action." (Solikyl volunteer)

Ultimately, the community decided to forgo adopting a hierarchical model due to a number of factors. As previously detailed in 4.1.1 initial attempts to organise into working groups proved unsuccessful which contributed to many food-sharers shying away from adopting a structured approach. Moreover, many food-sharers felt this would be unnecessary, at least during this early stage of the organisation. Although, food-sharers did explore the possibility for individual freedoms within the group. For example, in an attempt to find

balance between responsibility and control, they agreed on having a rotating facilitator for the weekly volunteer meetings, someone who would run the meeting by taking attendance, formulate an agenda based on attendees' suggestions, and make sure that everyone would have an opportunity to speak. And a 'temperature check' system was implemented as a way to quickly reach consensus on decisions where food-sharers would use hand gestures that were pre-agreed upon to communicate their feelings with regards to a particular decision – e.g., holding one's hand upwards and wiggling one's fingers to show agreement with a suggestion. Moreover, the Karrot platform included features that helped to facilitate this kind decentralised decision making. For instance, the community made use of a built in trust system that allows food-sharers to give and receive 'trust-karrots' to others, with more karrots equating to increased editing capabilities. It was also discussed that Karrot also has a built in voting system that would enable food-sharers to resolve any conflicts in a communal way, however this feature has not presently been used.

Outside of meetings, one food-sharer suggested that perhaps each individual could use their best judgement on making decisions that are time-sensitive or seem like they would be agreeable. This would practically entail that food-sharers could just act when needed, and then discuss with the whole group, when the first opportunity arises. This process would still allow discussions about wrong or right decisions, and still help the community to learn by trial and error and overtime develop a set of shared norms and values on which future decisions can be made (Tandon, 1997). Posting about the community in a local dumpster diving group on Facebook was regarded as an example of an 'easy' decision to which a food-sharer could 'just say yes', while establishing a partnership with a new food donor was probably something that should be discussed as a group. The idea was that this working strategy could result in more efficiency, in terms of making things happen, and reduce the decision fatigue among members.

However, as the community developed, it became more difficult to recognise what decisions that could be taken independently, and which ones require discussion. One food-sharer, for instance, shared an episode when she posted something on the community Facebook page that another contested as irrelevant. Relatedly, others have shown frustration with the flat hierarchy, and how to navigate it. Situations such as this highlight the messiness of grassroots democracy and how previously established norms can be subject to change. Being committed to a non-hierarchical decision-making structure has often been in tension with the desire to take practical action (Dana et al., 2021) – i.e., picking up food and redistributing it. Several food-sharers lamented that they experienced the initiative as 'all talk and no action', and some even felt that the community had lost potential members who were enthusiastic to get involved, but ultimately left as they were not interested in the more organisational, time-consuming aspects of the community. Periods of less structured decision-making were prompted by a feeling that the process was getting in the way of progress. As one food-sharer (volunteer) stated during an informal

conversation, *"the speed of action should be linked to the level of importance"* (Food-sharer 3).

4.3.2 Participation in flux and the challenges with governance

The decision to deviate from the suggestions of Solikyl and observed practices of FoodSharing Copenhagen and foodsharing.de, and adopt a non-hierarchical governance model became problematic with regards to fluctuating numbers of food-sharers, and their level of engagement with the community. During periods when food-sharers' participation is high (in this case, between twenty and thirty approximately), a more formal approach, that is reaching consensus on every decision at face-to-face meetings, was considered as necessary. During periods of low participation (approximately five to ten), the community found that disagreements rarely arose, so as a response a more pragmatic approach to governance was adopted. Decision making was moved online to Karrot, where issues could be shared as posts if and when they arose. This way food-sharers could simply make their feelings and opinions known in the comments. In practice, however, posts on Karrot tend to receive minimal interaction, due to varying levels of participation, but also to many food-sharers' lack of interest in organisational matters. Unlike the real-time deliberation that occurs during face-to-face meetings, discussions online were more fragmented and waiting for everybody to respond to a post proved to be impractical. Therefore, the community explicitly decided to use a combination of the lazy consensus model (Dana et al., 2021; Schneider, 2020) (where proposals within a group may be of presumed to pass unless any explicit objections arise) and a do-ocracy methodology [ibid]. This means that those who take the initiative to do work in a group are empowered to make decisions about what they do. This pragmatic solution allows food-sharers to take more of a backseat role with regards to the organisational work of the community if they so wish, but discussions and decisions are still visible to all who are signed up to the FoodSharing Stockholm group on the Karrot Platform. In a sense, the non-hierarchical model is framed and (re)interpreted within the practices of the community, where active participation and responsibilities define core food-sharers and their ability to make decisions. Ideals of democracy are practically redefined by volunteers' degree of participation and involvement in the community.

5 Discussion

CSCW scholarship on grassroots initiatives has illustrated how communities self-organise, and how digital technologies are appropriated to both structure key practices and contend with specific matters of concern (Berns & Rossitto, 2019; Berns et al., 2021a; Engelbutzeder et al., 2020; Ganglbauer et al., 2014; Lu, 2021; Mosconi et al., 2017). Zooming in the specific case of FoodSharing Stockholm, our analysis has unpacked the processes and everyday actions whereby a small grassroots initiative develops key visions and ways of organising collaborative work, from the ground up. Resonating with other CSCW

studies (Lampinen et al., 2022; Rossitto, 2021), our analysis has illustrated how processes of setting up, infrastructuring and governing the community develop through a set of inter-connections with related initiatives, as narratives about impact, key values and practices, ways of organising food distribution, along with the strategies to deal with various challenges are considered by members who have direct or indirect experience of similar initiatives. Processes of learning from other communities are of course not always linear and straightforward. We have seen, for instance, that a careful reflection of existing food-sharing models was instrumental to shape the community's own food distribution, while having to juggle with limited resources and potential problems. On the contrary, practices and models of governance were first adopted following the example of another community (Section 4.3.1), but then revised once core volunteers realised that processes of decision-making are highly influenced by the fluctuating number of central and more peripheral food-sharers (Section 4.3.1).

Furthermore, our analysis has shown that volunteers' awareness and previous experience of other initiatives provided a set of strategies to enrol volunteers and food donors, modalities and framings to participate in food distribution that could be appropriated, thus alleviating some of the problems of early organising. By learning from these similar food-sharing initiatives, the work of setting up FoodSharing Stockholm was significantly eased, as food-sharers could appropriate, or take as inspiration, existing participation and sharing modalities, had insights into common troubles encountered, that defining visions and key values was essential from the beginning , and that a structured organisation, with clear division of roles and tasks, requires a stable and large volunteer base.

We refer to this emerging body of previous experiences as *Collective histories of organising*. The term signifies the many subjective understandings, knowledge and awareness (e.g., what needs to be done and what strategies can help, challenges, problems to be avoided) that each volunteer brings to specific community-led initiatives, and the processes whereby these stories become contextually relevant to set up, sustain, and govern a community. We use the adjective collective to emphasise that, although previous experience can be highly individual, they gain (new) meanings in the situated interactions between community's members, and between community members, specific technologies, socio-cultural contexts and material environments. As communities' sociotechnical practices are envisioned and set up, or concrete problems addressed, experiences are put into circulation (in meetings, for instance) and discussed, oftentimes shaping each other. Their collective relevance can become manifest through planned meetings or happenstance, while sense-making processes (see (Mccarthy & Wright, 2004)), such as recounting, discussing or negotiating are central to how their relevance is framed.

Mobilising resources, organising communities and participating in them entail diverse collaborative partnerships between groups of actors (e.g., donors, food-sharers and food-savers), as well as the collaborative efforts needed to

make communities work. We call for analytical attention towards the the power relations (e.g., returning volunteers, claims donors might have, or even power structures embedded in the digital infrastructure) that might surface as different experiences are discussed and that could shape what histories become more relevant within a certain community.

"Collective histories of organising" emphasises the explicit and more tacit interconnections that community-led initiatives, striving for social change, develop with each other. This is relevant to CSCW research focused on understanding and designing for the volunteer-based work required to organise and maintain collective actions. We suggest that the notion can provide an alternative to concepts such as organisational memory, often used in relation to workplaces with a more formal organisational structure to emphasise processes of acquisition and management of existing relevant knowledge. Finally, "Collective histories of organising" emphasises the impact that previous experiences can have on communities' efforts to articulate their work. Articulation work (Star & Strauss, 1999) indicates the activities actors engage with to enact collaboration and division of labour, including aspects like communication, bringing attention to, planning and scheduling work. By re-orientating analytical and design efforts towards the *adaptability* and *adoptability* of existing sociotechnical practices that community-led initiatives can learn from each other, the notion helps understand the invisible aspects that shape community organising.

In line with the PAR approach of the presented study (Hayes, 2011, 2014; Kindon et al., 2007), and a concern for how community-led initiatives can benefit from research, we now discuss three themes that can help other communities to practically design sociotechnical practices for building, sustaining, and infrastructuring foodsharing initiatives. Previous work has provided guidelines on the role of queuing mechanisms at food-sharing events (Berns et al., 2021a, 2021b), and illustrated processes whereby food sharing communities develop and grow (Biørn-Hansen & Håkansson, 2018; Bødker, Korsgaard, Lyle, & Saad-Sulonen, 2016). Having detailed how the FoodSharing Stockholm community developed through learning's from other communities, in the discussion we expand past work and continue the cycle by highlighting what other initiatives can learn from FoodSharing Stockholm. To do this, we draw attention to analytical and practical sensitivities to organise work and set up community activism from the ground up. Addressing the limited human, physical, economic, and infrastructural resources of volunteer-led communities, we point to the relevance of existing framings and sociotechnical practices in helping community-led initiatives get started and sustain their work and commitment to it. We unpack configuring capacities, sociotechnical practices, and participation as key aspects that activists and designers should consider in mobilising resources, organising, and more broadly designing for food-sharing communities. We connect to scholarship that conceives of design as a set of diffuse, non-expert practices for everyday problem-solving (Manzini & Coad, 2015).

5.1 Configuring capacities

As in the case of many social movements, resource mobilisation (Martin et al., 2015) has been a core aspect in building FoodSharing Stockholm. Resonating with other studies, this has included enrolling volunteers and gathering material resources (see also (Biørn-Hansen & Håkansson, 2018)), putting together a technological infrastructure (see also (Bødker, Korsgaard, & Saad-Sulonen, 2016)), and more broadly identifying and developing the different capacities that are needed to operate. For instance, formal and informal relationships between core community members, between community members and other stakeholders (e.g., food donors, managers of community centres), knowledge and context-specific practices (e.g., the different food-sharing models discussed in 4.3), physical infrastructures (e.g., locales to host sharing events), and the negotiation of norms and values (how to govern the community) can be seen as different instances to identify, negotiate, and create the resources that are needed to operate.

Unsurprisingly, enrolling food-sharers is a salient concern as they do the groundwork of coordinating with food-donors, organising events, transporting and sorting out food. Equally important are processes to attract food-donors, hosting locations for food distribution, and envisioning possible food-savers (recipients). While the creation of this network of actors could be imagined as a sequence of well-defined steps, the reality of organising is not so tidy. As outlined in the analysis, it is an alternation of pre-defined strategies, adjustments to emerging problems, and reconsideration of original plans. We have seen, for instance, that despite being strategically planned to develop diverse professional relationships, issues with enrolling food-donors and finding a locale for food distribution were eventually solved through some of the volunteers' personal connections. In our case, efforts to enrol donors included strategies to develop stable relationships with local businesses; this can be challenging as potential donors might already be involved in other collaborations, be unaware of food-sharing initiatives and how they work, or simply not interested in becoming involved. Initial strategies to enrol donors involved developing a script and list of Q&As that key volunteers could use to clearly communicate the goals and benefits of the organisation to potential donors. However, in the end harnessing pre-existing social ties proved more fruitful.

We suggest that processes aimed at forming such partnerships could first develop around already established local ties and personal relationships. As seen in the analysis, in the early moments of organising, providing potential partners with consistent information on food-waste reduction, the benefits of participation, and on the motivations and modalities to participate, is central to coordinate the work of the different volunteers and to more effectively enrol new donors. Once grassroots initiatives become more visible or grow, more formal roles or bespoke platforms for creating collaborations could be developed. Learning from other initiatives (Rossitto, Comber, Tholander, & Jacobsson, 2022), this could entail the work of ambassadors to promote the adoption of

environmentally concerned actions, the design of bespoke platforms to connect small initiatives promoting sustainable food consumption (Lampinen et al., 2022), or a list of best practices to more easily activate collective efforts. For instance, the website foodsaving.world, created by members of the German food-sharing community, helped to get FoodSharing Stockholm up and running by offering advice on how to enrol donors. This practice could be expanded to a more participatory forum, where multiple communities could share diverse insights and advice to those starting anew. As shown elsewhere (Lampinen et al., 2022), this can facilitate the circulation of key practices and values, and yet allow for the flexibility needed to tailor existing sociotechnical practices to new contexts.

Relatedly, our analysis has illustrated that developing the collaborations needed to operate is interwoven with the articulation of core food-sharing visions (e.g., charity, sustainability), and models (e.g., face-to-face sharing events, solidarity fridge, technology-mediated one-to-one exchange). Here, the volunteers' previous experiences with other food-sharing initiatives were central to considerations about food sharing as a potential practice of sustainability. As it was discussed, a charity initiative in this context would rely on a more stable and regular flow of food, which made other possible reasons for distributing surplus food (i.e., helping those in need) unfeasible. *Configuring capacities and defining visions are interwoven processes, and we invite members of grassroots initiatives to reflect on how – or the extent to which – they mutually shape each other.* For instance, in contexts where community-led food-sharing initiatives are not common, there might be biases about the idea that second-hand food is distributed to vulnerable people, and careful considerations should be made on how this narrative could be avoided or re-framed.

Related to processes of building capacities were also discussions on what size the community should aspire for. Knowledge about initiatives, such as FoodSharing Copenhagen, contributed to the awareness that a large size presupposes a structured organisation, for instance to enrol volunteers, introduce them to the doings of the community, and plan working shifts. In our case, the initial plans to have different working groups were difficult to uphold as the number of interested people dropped already after the first weeks, and concerns emerged about operating with limited resources. This developed into an explicit strategy to keep the community small –and thus manageable– to the extent that core food-sharers declined an interview at a national radio to avoid attracting more attention than they could handle (e.g., to many food-savers than they could handle at the distribution events) (Rossitto et al., 2020). As noted elsewhere (Berns et al., 2021a), there are tensions here about concerns for visibility, impact, and being able to carry out practical work. Previous work (Lampinen, Rossitto, & Gradin Franzén, 2019; Rossitto & Lampinen, 2018) has shown that too much visibility and rapid growth can be overwhelming for volunteer-led initiatives for which social ties and supportive relationships are central to their organisation. Consequently, as a practical guideline, *we see a*

need to carefully consider processes of growth, either by devising strategies to implement them (e.g., when? how? with which capacity?) or even questioning the approach as an obvious objective to aim for. Considerations on how to increase impact by activating related initiatives might be more useful than utilising digital technologies to promote growth in number.

5.2 Configuring sociotechnical practices

One specific aspect of developing capacities relates to the community's efforts to use digital technologies to document meetings, create shared information spaces, and gain visibility. Similarly to other studies on food initiatives (Berns et al., 2021a; Ganglbauer et al., 2014), an ecology of digital artefacts forms the digital infrastructure that FoodSharing Stockholm relies on for its day-to-day operations. Social media platforms are used by volunteers to advertise food distribution to broad audiences. Overtime, the private message functions of these platforms have also become a way, for interested people, to inquire about the community. While the immediate reaction to this was to respond to this interest through personal messages, food-sharers, who are responsible for managing the community's social media, became frustrated at spending time with potential participants when their curiosity did not result in concrete actions to take part. Our analysis has provided a concrete example to address this problem with an auto-reply, containing information about the community and an invitation to learn more by participating to a physical event, being sent out to anyone inquiring about the community. *Outlining constraints caused by limited time and other resources – and not merely concerns for visibility – we emphasise the need to develop sources of information that potential new food-sharers, food-savers and food-donors can access, both online (e.g., dedicated websites, forums, and social media groups) and offline (e.g., pamphlets, posters).*

Because of limited budget and economic constraints, community-led initiatives often adopt existing technologies and social media to organise their work (Costanza-Chock, 2018; Rossitto, Korsgaard, et al., 2021). FoodSharing Stockholm sets a different example by adopting Karrot, a dedicated platform purposefully designed to support collaboration between members of food-sharing communities. As seen in the analysis, Karrot provides support for decision-making, planning events, and store information about volunteers and food donors. The design of the platform reflects a decentralised governance model, as users of the platform can collectively entrust others to edit online settings. Nevertheless, despite nurturing values such trust and horizontal participation, we have seen that moving decision-making to the platform did not resolve issues with more active involvement, as not all volunteers might share the same interest to participate. Moreover, while FoodSharing Stockholm's ethos is characterised by a sense of openness towards newcomers, core volunteers have been uncomfortable with the idea of sharing information on Karrot with participants who eventually do not commit to a longer-term participation, as this might create unforeseen problems. For instance, through participation in another food community, the first author was aware of cases where people

have shown up to collect food from donors while pretending to be volunteers. Although this might be a rare incident, the episode reflects concerns to devise practices of access to main digital platforms, and to define which information should be available to broader audiences or limited to returning and trusted food-sharers. *We invite members of community-led initiatives to consider how the design of sociotechnical practices can enable pathways to information and different levels of participation.*. The "Reader-to-Leader Framework" (Preece & Shneiderman, 2009) that captures the journey of how people can become progressively more involved in online communities (i.e., reader, contributor, collaborator, leader) could be of inspiration here. However, we see the challenges of formally structuring different degrees of participation (see (Rossitto, Korsgaard, et al., 2021) on this point) for communities aspiring to symmetrical models of governance. Examples of how other volunteer-led communities utilise their website to describe key roles, working structures and groups (see (Berns & Rossitto, 2019) or (Hoffice, 2022)) can be useful to make explicit what is expected from participants without imposing (any transitions between) formal roles on them. This point is further addressed in the following section.

Resonating with other work (Bødker, Korsgaard, & Saad-Sulonen, 2016; Rossitto, Bogdan, & Severinson-Eklundh, 2014), the analysis has shown that some digital tools are collectively negotiated and some others introduced by persons who have previous experience with them. In our case, creating a digital infrastructure has resulted in a mix of open source and commercial platforms being used. Although more in line with the overall non-profit framing of the community, open source technologies such as the Karrot Platform, are not always optimal for reaching large audiences. Digital technologies such as social media provide templates for participation that are broadly and readily available, and easily adopted. This should, however, raise concerns about the structures and relations they reproduce, enforce, or maybe exclude. As seen in other food-sharing contexts, the adoption of specific platforms can contribute to a service-based approach to food sharing (Ganglbauer et al., 2014; Garthwaite, 2016; Vyas, Snow, & Mallett, 2015), and even hinder caring relationships (e.g., connecting to other people) that volunteers might value (Berns et al., 2021b; Rossitto, Korsgaard, et al., 2021). As a practical recommendation, *we suggest that communities should strategically map the variety of digital technologies used, what key functions and roles they provide, and who benefits from them. Equally important is the need to understand the value tension stemming from their use, and how they relate to core goals, visions, and work organisation.*

5.3 Configuring participation

Within FoodSharing Stockholm, relationships between food-sharers and food-donors are the only form of collaboration that is clearly defined. For other participant groups there are no clear-cut definitions of what being a member entails, apart from committing to contributing to the community. Food-savers participate on a flexible basis as there is no sign up or RSVP required for

collecting food, as is the case in other communities (for example (Ganglbauer et al., 2014)). And for food-sharers participation is also flexible, making it possible for volunteers to engage as much as they wish, and when they have time. In reality, however, the openness of this approach makes it difficult for interested people to join, in that there are no stable guidelines or clear routines to guide them through the community, and start contributing to it. This relates to Freeman's argument that: *"For everyone to have the opportunity to be involved in a given group and to participate in its activities the structure must be explicit, not implicit. The rules of decision-making must be open and available to everyone, and this can happen only if they are formalised"* (Freeman, 1972, p. 3). As the community is not registered as an official (non-profit) organisation, there has not been a push to regulate the community with clear guidelines and decision-making models. Anytime the topic of 'becoming official' has been raised, it has been dismissed based on concerns for the flexibility of the community, and too much bureaucracy. *Based on our analysis, we can however argue that making key rules, norms, and tasks more explicit could shape different actors' motivations and expectations about participation, and we invite grassroots initiatives to do so.*

In the analysis, we have discussed how balancing structure with pragmatism, and deciding what decision-making models to adopt were central governance issues. As we have seen, core food-sharers strive to maintain a non-hierarchical model for decision-making, and they encourage new members to take independent initiatives. However, this does not work well in practice, with newcomers being hesitant about making decisions, and turning to other food-sharers for advice. Moreover, in reality, structures always exist or emerge, whether formalised or not. A primary example within the community is what has become to be known as the 'core group' of food-sharers , who are considered instrumental in keeping the community alive even during periods of low participation. As an undesired consequence, this group has, overtime, become an informal hierarchical leadership structure. This has led to stress and feelings of obligation within the group, as none of these food-sharers have actively chosen to take on this role. It appears that the lack of organisational structure within the community prevents newcomers to take ownership of the project as there were no clear structures for how this should be done.

Our last recommendation, concerns sensitivities for organising participation and governance of volunteer-led communities. As members juggle with the conditions of flexible forms of participation and the practical concerns to distribute workload and responsibilities, there is a need for a governance model that can work with many, but also few people. Resonating with Freeman (Freeman, 1972), this entails effort to *make basic decision-making mechanisms visible, along with pre-defined and emerging structures.* For instance, similar to the 'trust-karrots' mechanism on the Karrot platform, decision-making could be rendered visible and made explicit through technology with the design of a community voting mechanism. Likewise, as a practical guideline, we suggest that, *rather than shying away from any formal structure, communities*

might try to recognise and acknowledge any informal structures that emerge naturally, and work to develop them into governance mechanisms that serve the community with regards to disturbing control and being adaptable to fluctuations in participation levels. For instance, in our case the "core group" of food-sharers could evolve into a 'self appointed board' (Dana et al., 2021) and, from there, strive for distribution of power on a rotating basis where existing board members collectively decide when to appoint new board members and who those members will be.

6 Conclusion

Past CSCW scholarship has investigated the role digital technology plays in structuring social movements and their efforts to organise collective action. However, relatively little was known everyday practical work whereby such initiates are formed, get started, and become communities. This paper expands previous work that has focused on established food-sharing communities, by exploring the organisational and collaborative work that goes into setting up such initiatives. The findings have unpacked how the FoodSharing Stockholm community developed through learning from other communities and the explicit and tacit interconnections that it developed with related, existing initiatives.In the discussion, we introduced the concept of *"Collective histories of organising"* to account for the many and diverse previous experiences that volunteers bring to processes of setting up and maintaining a community-led initiative. Finally, to continue the cycle of learning, we presented three themes that can help other grassroots communities to practically build, sustain, and infrastructure collective action by specifically addressing issues connected to configuring capacities, sociotechnical practices, and sharing mechanisms.

We encourage communities to define and make visible decision-making mechanisms, pre-defined and emerging structures. We also suggest that rather than avoiding formal structure, communities should recognise and acknowledge any informal structures that emerge naturally, and work to develop them into governance mechanisms that serve the community. We encourage researchers and activists to consider how grassroots, community-led initiatives develop through the set of inter-connections, outcomes, and narratives about impact circulate locally, and how key practices, ways of organising, and modes of participation are adopted across contexts. Recognising how grassroots communities vary in terms of capacities, sociotechnical practices and forms of participation we are not suggesting universal, abstract solutions for organising collective action around food waste. Rather, we hope that researchers, designers and activists will reflect on these overarching themes within specific food-sharing settings, and will continue evolving this research and activist space with specific instances of both success stories and challenges of organising that other communities can learn from.

7 Declarations

7.1 Ethical Approval

Not applicable, ethical approval was not required for this study.

7.2 Competing interests

The first author was a member of the FoodSharing Stockholm community during the data collection and analysis. The nature of this relationship in described in detail in section 3.3 (Reflections on ethics and positionality).

7.3 Authors' contributions

The data collection and analysis were carried out by the first author, K.B. The manuscript was written by all three authors and K.B prepared the figure. The manuscript was reviewed by K.B and C.R.

7.4 Funding

The study was carried out as part of the first authors PhD project at Stockholm University, no external funding was obtained.

7.5 Availability of data and materials

The data-sets used can be accessed by contacting the corresponding author at katie@dsv.su.se.

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References

Altarriba, F., Lanzani, S.E., Torralba, A., Funk, M. (2017, June). The Grumpy Bin: Reducing Food Waste Through Playful Social Interactions. *Proceedings of the 2017 ACM Conference Companion Publication on Designing Interactive Systems* (pp. 90–94). New York, NY, USA: Association for Computing Machinery. 10.1145/3064857.3079125

Aschemann-Witzel, J., De Hooge, I., Amani, P., Bech-Larsen, T., Oostindjer, M. (2015). Consumer-related food waste: Causes and potential for action. *Sustainability*, *7*(6), 6457–6477.

Baumer, E.P., Xu, X., Chu, C., Guha, S., Gay, G.K. (2017, February). When Subjects Interpret the Data: Social Media Non-use as a Case for

Adapting the Delphi Method to CSCW. *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing* (pp. 1527–1543). New York, NY, USA: Association for Computing Machinery. 10.1145/2998181.2998182

Berns, K., & Rossitto, C. (2019). From Commodities to Gifts: Redistributing Surplus Food Locally. *Ethnographies of Collaborative Economies, Edinburgh, Scotland, 25 October, 2019* (p. 11). Edinburgh, Scotland: Sharing & Caring - COST Action CA16121.

Berns, K., Rossitto, C., Tholander, J. (2021a, January). Queuing for Waste: Sociotechnical Interactions within a Food Sharing Community. *CHI '21 Making Waves, Combining Strengths* (p. 10). Online Virtual Conference (originally Yokohama, Japan): Association for Computing Machinery. 10.1145/3411764.3445059

Berns, K., Rossitto, C., Tholander, J. (2021b, June). "This is not a free supermarket": Reconsidering Queuing at Food-sharing Events. *C&T '21: Proceedings of the 10th International Conference on Communities & Technologies - Wicked Problems in the Age of Tech* (pp. 319–331). New York, NY, USA: Association for Computing Machinery. 10.1145/3461564.3461582

Biørn-Hansen, A., & Håkansson, M. (2018, April). Building Momentum: Scaling up Change in Community Organizations. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). New York, NY, USA: Association for Computing Machinery. 10.1145/3173574.3173984

Braun, V., & Clarke, V. (2012). Thematic analysis. *APA handbook of research methods in psychology.* (Vol. Vol. 2, pp. 57–71). 750 First Street NE, Washington: American Psychological Association. (Publisher: American Psychological Association)

Brown, B., Weilenmann, A., McMillan, D., Lampinen, A. (2016, May). Five Provocations for Ethical HCI Research. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (pp. 852–863). New York, NY, USA: Association for Computing Machinery. 10.1145/2858036.2858313

Brünker, F., Deitelhoff, F., Mirbabaie, M. (2019, December). *Collective Identity Formation on Instagram – Investigating the Social Movement Fridays for Future.* arXiv.

Burton, E., Meier, C., Olarte, R., Skeini, H., Zahan, F. (2017, November). Airshare: a food sharing concept. *Proceedings of the 29th Australian Conference on Computer-Human Interaction* (pp. 634–639). New York, NY,

USA: Association for Computing Machinery. 10.1145/3152771.3154859

Butzlaff, F., & Deflorian, M. (2021). Direct social action beyond party politics. How new subjectivities change the idea of social transformation. *Political Research Exchange*, *3*(1), 1896343.

Bødker, S., Korsgaard, H., Lyle, P., Saad-Sulonen, J. (2016, October). Happenstance, Strategies and Tactics: Intrinsic Design in a Volunteer-based Community. *Proceedings of the 9th Nordic Conference on Human-Computer Interaction* (pp. 1–10). New York, NY, USA: Association for Computing Machinery. 10.1145/2971485.2971564

Bødker, S., Korsgaard, H., Saad-Sulonen, J. (2016, February). 'A Farmer, a Place and at least 20 Members': The Development of Artifact Ecologies in Volunteer-based Communities. *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing* (pp. 1142–1156). New York, NY, USA: Association for Computing Machinery. 10.1145/2818048.2820029

Cherry, M.A. (2016, March). *Legal and Governance Structures Built to Share* (SSRN Scholarly Paper No. 2748865). Rochester, NY: Social Science Research Network.

Chies, B.M. (2017). Turning Food 'waste' into a Commons. *Practicing the Commons. Self-Governance, Cooperation and Institutional Change. XVI Biennal IASC-Conference. https://www. iasc2017. org/wp-content/uploads/2017/07/chies. pdf* (p. 17). Utrecht: Digital Library on the Commons.

Ciaghi, A., & Villafiorita, A. (2016, September). Beyond food sharing: Supporting food waste reduction with ICTs. *2016 IEEE International Smart Cities Conference (ISC2)* (pp. 1–6). Trento, Italy: IEEE. 10.1109/ISC2.2016.7580874

Comber, R., Hoonhout, J., van Halteren, A., Moynihan, P., Olivier, P. (2013, April). Food practices as situated action: exploring and designing for everyday food practices with households. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 2457–2466). New York, NY, USA: Association for Computing Machinery. 10.1145/2470654.2481340

Costanza-Chock, S. (2018). *Design Justice: Towards an Intersectional Feminist Framework for Design Theory and Practice*. Rochester, NY: Social Science Research Network.

Dahlberg-Grundberg, M. (2016, October). Technology as movement: On hybrid organizational types and the mutual constitution of movement identity and technological infrastructure in digital activism. *Convergence*, *22*(5), 524–542.

10.1177/1354856515577921

Dana, C., Hornbein, D., Russell, V., Schneider, N. (2021). *Community Rules, Simple Templates for Great Communities*. University of Colorado Boulder: Media Enterprise Design Lab.

Dantec, C.A.L. (2016). *Designing Publics*. Cambridge, Massachusetts, United States: MIT Press.

Davies, A. (2019). *Urban Food Sharing: Rules, Tools and Networks*. Ireland: Policy Press.

Davies, A., Edwards, F., Marovelli, B., Morrow, O., Rut, M., Weymes, M. (2017, April). Creative construction: Crafting, negotiating and performing urban food sharing landscapes. *Area*, *49*, 9.

10.1111/area.12340

Davies, A.R. (2009). Does sustainability count? Environmental policy, sustainable development and the governance of grassroots sustainability enterprise in Ireland. *Sustainable Development*, *17*(3), 174–182.

10.1002/sd.374

Davies, A.R., & Legg, R. (2018). Fare sharing: interrogating the nexus of ICT, urban food sharing, and sustainability. *Food, Culture & Society*, *21*(2), 233–254.

DeWalt, K.M., & DeWalt, B.R. (2002). *Participant Observation: A Guide for Fieldworkers*. Walnut Creek, CA: Rowman Altamira.

Dewey, J. (2012). *The Public and Its Problems: An Essay in Political Inquiry*. Pennsylvania: Penn State University Press.

Dickinson, J., Díaz, M., Le Dantec, C.A., Erete, S. (2019, November). "The cavalry ain't coming in to save us": Supporting Capacities and Relationships through Civic Tech. *Proceedings of the ACM on Human-Computer Interaction*, *3*(CSCW), 123:1–123:21.

10.1145/3359225

DiSalvo, C. (2009, January). Design and the Construction of Publics. *Design Issues*, *25*(1), 48–63.

10.1162/desi.2009.25.1.48

Dombrowski, L., Brubaker, J.R., Hirano, S.H., Mazmanian, M., Hayes, G.R. (2013, September). It takes a network to get dinner: designing location-based systems to address local food needs. *Proceedings of the 2013 ACM international joint conference on Pervasive and ubiquitous computing* (pp. 519–528). New York, NY, USA: Association for Computing Machinery. 10.1145/2493432.2493493

Dosono, B., & Semaan, B. (2019, May). Moderation Practices as Emotional Labor in Sustaining Online Communities: The Case of AAPI Identity Work on Reddit. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). New York, NY, USA: Association for Computing Machinery. 10.1145/3290605.3300372

Earl, J., & Kimport, K. (2011). *Digitally Enabled Social Change: Activism in the Internet Age.* Boston: The MIT Press.

Engelbutzeder, P., Cerna, K., Randall, D., Lawo, D., Müller, C., Stevens, G., Wulf, V. (2020, October). Investigating the use of digital artifacts in a community project of sustainable food practices: ‘My chili blossoms’. *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society* (pp. 1–4). New York, NY, USA: Association for Computing Machinery.

Erete, S.L. (2015, February). Engaging Around Neighborhood Issues: How Online Communication Affects Offline Behavior. *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing* (pp. 1590–1601). New York, NY, USA: Association for Computing Machinery. 10.1145/2675133.2675182

fairteiler, F. (2022, August). *Lebensmittel teilen, statt wegwerfen - foodsharing Deutschland.* Retrieved 2022-08-08, from https://foodsharing.de

Falcone, P.M., & Imbert, E. (2017). Bringing a Sharing Economy Approach into the Food Sector: The Potential of Food Sharing for Reducing Food Waste. P. Morone, F. Papendiek, & V.E. Tartiu (Eds.), *Food Waste Reduction and Valorisation: Sustainability Assessment and Policy Analysis* (pp. 197–214). Cham: Springer International Publishing. 10.1007/978-3-319-50088-1_10

Farr-Wharton, G., Choi, J.H.-J., Foth, M. (2014, December). Food talks back: exploring the role of mobile applications in reducing domestic food wastage. *Proceedings of the 26th Australian Computer-Human*

Interaction Conference on Designing Futures: the Future of Design (pp. 352–361). New York, NY, USA: Association for Computing Machinery. 10.1145/2686612.2686665

Feng, Y., Marek, C., Tosun, J. (2022). Fighting Food Waste by Law: Making Sense of the Chinese Approach. *Journal of Consumer Policy*, *45*(3), 457–479.

foodsaving.world (2022). *Foodsaving Worldwide | FSWW.* Retrieved 2022-11-23, from https://foodsaving.world/

foodsharingcph.org (2020). *Foodsharingcph.org.* Retrieved 2020-05-04, from https://foodsharingcph.org/ (Library Catalog: foodsharingcph.org)

foodsharing.de (2020). *Lebensmittel teilen, statt wegwerfen - foodsharing Deutschland.* Retrieved 2020-05-04, from https://foodsharing.de (Library Catalog: foodsharing.de)

Freeman, J. (1972). The Tyranny Of Structurelessness. *Berkeley Journal of Sociology*, *17*, 151–164.

Ganglbauer, E., Fitzpatrick, G., Comber, R. (2013, May). Negotiating food waste: Using a practice lens to inform design. *ACM Transactions on Computer-Human Interaction*, *20*(2), 11:1–11:25.

10.1145/2463579.2463582

Ganglbauer, E., Fitzpatrick, G., Molzer, G. (2012, December). Creating visibility: understanding the design space for food waste. *Proceedings of the 11th International Conference on Mobile and Ubiquitous Multimedia* (pp. 1–10). New York, NY, USA: Association for Computing Machinery. 10.1145/2406367.2406369

Ganglbauer, E., Fitzpatrick, G., Subasi, O., Güldenpfennig, F. (2014, February). Think globally, act locally: a case study of a free food sharing community and social networking. *Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing* (pp. 911–921). New York, NY, USA: Association for Computing Machinery. 10.1145/2531602.2531664

Garthwaite, K. (2016). Stigma, shame and'people like us': an ethnographic study of foodbank use in the UK. *Journal of Poverty and Social Justice*, *24*(3), 277–289.

Ghoshal, S., Parker, A.G., Le Dantec, C.A., Disalvo, C., Irani, L., Bruckman, A. (2019, November). Design and the Politics of Collaboration: A Grassroots Perspective. *Conference Companion Publication of the 2019 on Computer Supported Cooperative Work and Social Computing* (pp. 468–473). New York, NY, USA: Association for Computing Machinery. 10.1145/3311957.3359438

Gibson-Graham, Healy, S., Cameron, J. (2016). *Commoning as a postcapitalist politics 1.* 2 Park Square, Milton Park, Abingdon, Oxon OX14 4RN: Routledge.

Gollnhofer, J.F., & Boller, D. (2020). The Evolution of the German Anti-food Waste Movement: Turning Sustainable Ideas into Business. E. Narvanen, N. Mesiranta, M. Mattila, & A. Heikkinen (Eds.), *Food Waste Management: Solving the Wicked Problem* (pp. 115–139). Cham: Springer International Publishing.

Gustavsson, J., Cederberg, C., Sonesson, U. (2011). *Global food losses and food waste: extent, causes and prevention ; study conducted for the International Congress Save Food! at Interpack 2011, [16 - 17 May], Düsseldorf, Germany.* Rome: Food and Agriculture Organization of the United Nations.

Hansson, K., Pargman, T.C., Bardzell, S. (2021, December). Materializing activism. *Computer Supported Cooperative Work (CSCW)*, *30*(5), 617–626.

10.1007/s10606-021-09412-5

Hansson, K., Sveningsson, M., Ganetz, H. (2021, December). Organizing Safe Spaces: #MeToo Activism in Sweden. *Computer Supported Cooperative Work (CSCW)*, *30*(5), 651–682.

10.1007/s10606-021-09410-7

Hayes, G.R. (2011). The relationship of action research to human-computer interaction. *ACM Transactions on Computer-Human Interaction (TOCHI)*, *18*(3), 1–20.

Hayes, G.R. (2014). Knowing by doing: action research as an approach to HCI. *Ways of Knowing in HCI* (pp. 49–68). New York, NY, USA: Springer.

Heitlinger, S., Bryan-Kinns, N., Comber, R. (2018, August). Connected seeds and sensors: co-designing internet of things for sustainable smart cities with urban food-growing communities. *Proceedings of the 15th Participatory Design Conference: Short Papers, Situated Actions, Workshops*

and Tutorial - Volume 2 (pp. 1–5). New York, NY, USA: Association for Computing Machinery. 10.1145/3210604.3210620

Heitlinger, S., Bryan-Kinns, N., Comber, R. (2019, May). The Right to the Sustainable Smart City. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–13). New York, NY, USA: Association for Computing Machinery. 10.1145/3290605.3300517

HLPE (2014). *Food losses and waste in the context of sustainable food systems: A report by the High Level Panel of Experts on Food Security and Nutrition of the Committee on World Food Security, Rome 2014.* (Tech. Rep.). Rome, Italy: FAO.

Hoffice (2022, August). *Hoffice | Come and work at someone's home.* Retrieved 2022-08-10, from http://hoffice.nu/en/

karrot.world (2022). *Karrot.*

Kera, D., & Sulaiman, N.L. (2014, October). FridgeMatch: Design probe into the future of urban food commensality. *Futures*, *62*, 194–201.

10.1016/j.futures.2014.04.007

Kindon, S., Pain, R., & Kesby, M. (Eds.). (2007). *Participatory Action Research Approaches and Methods: Connecting People, Participation and Place.* London: Routledge.

Klandermans, B., & Oegema, D. (1987). Potentials, networks, motivations, and barriers: Steps towards participation in social movements. *American sociological review*, *1*(1), 519–531.

Lampinen, A. (2021, August). The Trouble With Sharing. *Synthesis Lectures on Human-Centered Informatics*, *14*(4), i–103.

10.2200/S01116ED1V01Y202107HCI051

Lampinen, A., Light, A., Rossitto, C., Fedosov, A., Bassetti, C., Bernat, A., ... Avram, G. (2022, January). Processes of Proliferation: Impact Beyond Scaling in Sharing and Collaborative Economies. *Proceedings of the ACM on Human-Computer Interaction*, *6*(GROUP), 41:1–41:22.

10.1145/3492860

Lampinen, A., Rossitto, C., Gradin Franzén, C. (2019). Scaling Out, Scaling Down : Reconsidering growth in grassroots initiatives. *Ethnographies of Collaborative Economies, Edinburgh, Scotland, 25 October, 2019* (p. 10).

Edinburgh, Scotland, UK: Sharing & Caring - COST Action CA16121.

Lazell, J. (2016). Consumer food waste behaviour in universities: Sharing as a means of prevention. *Journal of Consumer Behaviour*, *15*(5), 430–439.

10.1002/cb.1581

Le Dantec, C.A., & DiSalvo, C. (2013). Infrastructuring and the formation of publics in participatory design. *Social Studies of Science*, *43*(2), 241–264. (Publisher: Sage Publications, Ltd.)

Le Dantec, C.A., Farrell, R.G., Christensen, J.E., Bailey, M., Ellis, J.B., Kellogg, W.A., Edwards, W.K. (2011, May). Publics in practice: ubiquitous computing at a shelter for homeless mothers. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 1687–1696). New York, NY, USA: Association for Computing Machinery. 10.1145/1978942.1979189

Light, A., & Miskelly, C. (2019, June). Platforms, Scales and Networks: Meshing a Local Sustainable Sharing Economy. *Computer Supported Cooperative Work*, *28*(3-4), 591–626.

10.1007/s10606-019-09352-1

Lu, K. (2021, October). Designing Democratic Systems for Civic Collective Action. *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing* (pp. 270–274). New York, NY, USA: Association for Computing Machinery. 10.1145/3462204.3481792

Manzini, E., & Coad, R. (2015). *Design, When Everybody Designs: An Introduction to Design for Social Innovation.* Cambridge, Massachusetts, United States: The MIT Press.

Martin, C.J., Upham, P., Budd, L. (2015, October). Commercial orientation in grassroots social innovation: Insights from the sharing economy. *Ecological Economics*, *118*, 240–251.

10.1016/j.ecolecon.2015.08.001

Mccarthy, J., & Wright, P. (2004, September). Technology as Experience. *Technology as Experience* (Vol. 11). (Journal Abbreviation: interactions) 10.1145/1015530.1015549

McCarthy, J., & Wright, P. (2015). *Taking [A]part: The Politics and Aesthetics of Participation in Experience-Centered Design.* Cambridge, MA, USA:

MIT Press.

Menendez-Blanco, M., Bjorn, P., De Angeli, A. (2017, February). Fostering Cooperative Activism through Critical Design. *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing* (pp. 618–629). New York, NY, USA: Association for Computing Machinery. 10.1145/2998181.2998198

Michelini, L., Principato, L., Iasevoli, G. (2018, March). Understanding Food Sharing Models to Tackle Sustainability Challenges. *Ecological Economics*, *145*, 205–217.

10.1016/j.ecolecon.2017.09.009

Mosconi, G., Korn, M., Reuter, C., Tolmie, P., Teli, M., Pipek, V. (2017, December). From Facebook to the Neighbourhood: Infrastructuring of Hybrid Community Engagement. *Computer Supported Cooperative Work (CSCW)*, *26*(4), 959–1003.

10.1007/s10606-017-9291-z

Mundt, M., Ross, K., Burnett, C.M. (2018, October). Scaling Social Movements Through Social Media: The Case of Black Lives Matter. *Social Media + Society*, *4*(4), 2056305118807911.

10.1177/2056305118807911

Preece, J., & Shneiderman, B. (2009, March). The Reader-to-Leader Framework: Motivating Technology-Mediated Social Participation. *AIS Transactions on Human-Computer Interaction*, *1*(1), 13–32.

Prost, S. (2019, October). Food Democracy for All? Developing a Food Hub in the Context of Socio-Economic Deprivation. *Politics and Governance*, *7*(4), 142–153.

10.17645/pag.v7i4.2057

Prost, S., Crivellaro, C., Haddon, A., Comber, R. (2018, April). Food Democracy in the Making: Designing with Local Food Networks. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (pp. 1–14). New York, NY, USA: Association for Computing Machinery. 10.1145/3173574.3173907

Rombach, M., & Bitsch, V. (2015). Food movements in Germany: slow food, food sharing, and dumpster diving. *International Food and Agribusiness Management Review*, *18*(1030-2016-83042), 1–24.

Rossitto, C. (2021, April). Political Ecologies of Participation: Reflecting on the Long-term Impact of Civic Projects. *Proceedings of the ACM on Human-Computer Interaction*, *5*(CSCW1), 187:1–187:27.

10.1145/3449286

Rossitto, C., Berns, K., Larsen-Ledet, I. (2020, October). Strategies of visibility: growth, media and social movements. *The 23rd ACM Conference on Computer-Supported Cooperative Work and Social Computing* (p. 5). New York, NY, USA: ACM.

Rossitto, C., Bogdan, C., Severinson-Eklundh, K. (2014, April). Understanding Constellations of Technologies in Use in a Collaborative Nomadic Setting. *Computer Supported Cooperative Work (CSCW)*, *23*(2), 137–161.

10.1007/s10606-013-9196-4

Rossitto, C., Comber, R., Tholander, J., Jacobsson, M. (2022, April). Towards Digital Environmental Stewardship: the Work of Caring for the Environment in Waste Management. *CHI Conference on Human Factors in Computing Systems* (pp. 1–16). New York, NY, USA: Association for Computing Machinery. 10.1145/3491102.3517679

Rossitto, C., Korsgaard, H., Lampinen, A., Bødker, S. (2021, October). Efficiency and Care in Community-led Initiatives. *Proceedings of the ACM on Human-Computer Interaction*, *5*(CSCW2), 467:1–467:27.

10.1145/3479611

Rossitto, C., & Lampinen, A. (2018, December). Co-Creating the Workplace: Participatory Efforts to Enable Individual Work at the Hoffice. *Computer Supported Cooperative Work (CSCW)*, *27*(3), 947–982.

10.1007/s10606-018-9319-z

Rossitto, C., Lampinen, A., Light, A., Diogo, V., Bernat, A. (2021). Why Are We Still Using Facebook? The Platform Paradox in Collaborative Community Initiatives. *BECOMING A PLATFORM IN EUROPE* (pp. 90–109). Boston - Delft: Now Publishers Inc.

Schanes, K., & Stagl, S. (2019, February). Food waste fighters: What motivates people to engage in food sharing? *Journal of Cleaner Production*, *211*, 1491–1501.

10.1016/j.jclepro.2018.11.162

Schmid, B. (2021). Hybrid infrastructures: The role of strategy and compromise in grassroot governance. *Environmental Policy and Governance*, *31*(3), 199–210.

10.1002/eet.1929

Schneider, N. (2020, December). *Designing Community Self-Governance with CommunityRule* (Tech. Rep.). USA: MediArXiv.

Schneider, N. (2021a, January). Admins, mods, and benevolent dictators for life: The implicit feudalism of online communities. *New Media & Society*, *1*(1), 1461444820986553.

10.1177/1461444820986553

Schneider, N. (2021b, February). The Tyranny of openness: what happened to peer production? *Feminist Media Studies*, *1*(1), 1–18.

10.1080/14680777.2021.1890183

Schrock, A. (2018). *Civic Tech: Making Technology Work for People.* Rogue Academic.

solikyl.se (2019). *Help us save food!* Retrieved 2021-02-26, from https://solikyl.se/

Star, S.L., & Strauss, A. (1999, March). Layers of Silence, Arenas of Voice: The Ecology of Visible and Invisible Work. *Computer Supported Cooperative Work (CSCW)*, *8*(1), 9–30.

10.1023/A:1008651105359

Svenfelt, A., & Zapico, J.L. (2016, August). Sustainable food systems with ICT. *Smart and Sustainable* (pp. 194–201). Amsterdam, the Netherlands: Atlantis Press. (ISSN: 2352-538X) 10.2991/ict4s-16.2016.23

Tandon, R. (1997). *Grassroots Democracy: Governance as If Citizens Mattered.* Society for Participatory Research in Aisa (PRIA).

Tartiu, V.E., & Morone, P. (2017). Grassroots Innovations and the Transition Towards Sustainability: Tackling the Food Waste Challenge. P. Morone, F. Papendiek, & V.E. Tartiu (Eds.), *Food Waste Reduction and Valorisation: Sustainability Assessment and Policy Analysis* (pp. 303–327). Cham: Springer International Publishing.

United Nations (2013). *THE 17 GOALS | Sustainable Development.* Retrieved 2021-02-26, from https://sdgs.un.org/goals

Voida, A., Yao, Z., Korn, M. (2015, February). (Infra)structures of Volunteering. *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing* (pp. 1704–1716). New York, NY, USA: Association for Computing Machinery. 10.1145/2675133.2675153

Vyas, D., Snow, S., Mallett, M. (2015, December). More than just Food: Field Visits to an Emergency Relief Centre. *Proceedings of the Annual Meeting of the Australian Special Interest Group for Computer Human Interaction* (pp. 662–666). New York, NY, USA: Association for Computing Machinery. 10.1145/2838739.2838787

Wilson, T., Zhou, K., Starbird, K. (2018, November). Assembling Strategic Narratives: Information Operations as Collaborative Work within an Online Community. *Proceedings of the ACM on Human-Computer Interaction*, *2*(CSCW), 183:1–183:26.

10.1145/3274452